

PML-AI Body of Knowledge

The reference for the PCI Project Management Leader – AI
Leadership · delivery systems · governance · the governed use
of AI

FIRST EDITION — PHASE 1 PROTOTYPE (NOT FOR RELEASE)

PROJECT CONTROLS INSTITUTE GLOBAL

Finance intelligently. Control predictively. Deliver successfully.

AI proposes; the professional verifies, decides and remains accountable.

— Prototype notice

This document is a **Phase 1 production prototype** of one domain, built to validate the book's pattern, calculation-verification harness, figures and typesetting pipeline before scaled authorship. It is not a released PCI publication and carries no entitlement, syllabus or examination status. The released edition will carry the full copyright, disclaimer and trademark notices of the PCI book family; all examples here are original and fictional, all calculations pass the golden-answer suite ([verify_formulas.py](#)), and no standard's text is reproduced.

Contents

The Project Leadership Profession	6
Why this domain exists	6
Knowledge Area 1.1 Projects, programmes and portfolios	7
Knowledge Area 1.2 The project leader's accountability	9
Knowledge Area 1.3 Systems thinking and value	12
Knowledge Area 1.4 Ethics and the responsible use of AI	15
Advanced topics — Domain 1	18
Industry variations — Domain 1	18
Case study — Domain 1: Meridian under scrutiny (public health)	19
Case study B — Domain 1: the plan nobody could critique (financial services)	19
Executive perspective — Domain 1	20
Calculation exercises — Domain 1	20
Practitioner's toolkit — Domain 1	20
Exam preparation — Domain 1	21
Domain 1 summary	21
Planning, Scheduling and Delivery Flow (quantitative flagship)	23
Why this domain exists	23
Knowledge Area 6.1 Planning levels and logic networks	24
Knowledge Area 6.2 The critical path and float	27
Knowledge Area 6.3 Resources, constraints, milestones and rolling wave	30
Knowledge Area 6.4 Delivery flow: predictive, agile and hybrid; recovery and forecasting	34
Advanced topics — Domain 6	39
Industry variations — Domain 6	40
Case study — Domain 6: recovering Auriga (utilities / technology)	40
Case study B — Domain 6: the milestone that was typed, not computed (public programme)	41
Executive perspective — Domain 6	41
Calculation exercises — Domain 6	42
Practitioner's toolkit — Domain 6	42
Exam preparation — Domain 6	43
Domain 6 summary	43
Cost, Resources and Commercial Awareness (quantitative flagship)	45
Why this domain exists	45
Knowledge Area 7.1 Estimating and budgeting	46
Knowledge Area 7.2 The cost baseline, actuals and forecasting	49
Knowledge Area 7.3 Earned value: measurement, variances and forecasting	50
Knowledge Area 7.4 Resource economics, procurement strategy and cash	55
Advanced topics — Domain 7	58
Industry variations — Domain 7	58
Case study — Domain 7: the forecast the board actually needed (utilities)	59
Case study B — Domain 7: past the point of total assumption (technology)	59
Executive perspective — Domain 7	60

Calculation exercises — Domain 7	60
Practitioner's toolkit — Domain 7	61
Exam preparation — Domain 7	62
Domain 7 summary	62
Risk, Uncertainty and Resilience (quantitative)	63
Why this domain exists	63
Knowledge Area 8.1 Threats, opportunities and identification	64
Knowledge Area 8.2 Analysis: from screening to quantification	66
Knowledge Area 8.3 Responses, reserves and governance	71
Knowledge Area 8.4 Resilience, bias and crisis leadership	73
Advanced topics — Domain 8	75
Industry variations — Domain 8	76
Case study — Domain 8: the survey Auriga did not commission (utilities)	76
Case study B — Domain 8: the diversified portfolio that wasn't (technology programme)	77
Executive perspective — Domain 8	77
Calculation exercises — Domain 8	78
Practitioner's toolkit — Domain 8	78
Exam preparation — Domain 8	79
Domain 8 summary	79

01

PART ONE

Leading projects

Domains 1-4 — the profession, strategy and selection, governance and decision rights, and delivery architecture: what a project leader is for and answerable for.



01

DOMAIN 1

The Project Leadership Profession

Group: Leading projects (Domain 1 of 4 in Part One). **Target:** ~68 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries ([docs/books/registries/](#)). This domain fixes the book's core vocabulary — accountability, outputs/outcomes/benefits, systems thinking, the responsible-AI principle — that every later domain assumes. British English; USD (+SAR where useful, indicative USD 1 ≈ SAR 3.75).

— Why this domain exists

Every technique in this book — the networks of Domain 6, the earned value of Domain 7, the risk quantification of Domain 8 — is instrumentation. This domain is about the person holding the instruments: what a project leader is for, what they are answerable for, and how they think. It establishes the delivery landscape and how leading temporary work differs from running a steady-state function (KA 1.1); defines accountability precisely, because the word is used loosely and the distinction decides who answers when things go wrong (KA 1.2); builds the two habits of mind that separate a leader from a coordinator — systems thinking, and the discipline of outputs versus outcomes versus benefits (KA 1.3); and grounds professional ethics and the governed use of AI (KA 1.4). A reader who finishes only this domain should already reason like the profession: know what success actually means, know who owns it, and stay answerable for the judgment even when a machine produced the analysis.

Learning objectives. After this domain a candidate can: distinguish projects, programmes and portfolios and explain what each optimises; contrast project leadership with operational leadership; explain why a temporary organisation makes trust and clarity urgent; distinguish accountability from responsibility and apply the distinction to a delegation; state the leader's obligations to sponsor, team, users and the public, and where they conflict; explain the professional standard of care; analyse a project as a system with feedback and delay; distinguish outputs, outcomes and benefits and demonstrate why output-based claims overstate value; compute the value of time in benefit terms; describe the ethical obligations of the role; and apply the PCI responsible-AI principle to a realistic delivery decision.

The master case. One programme runs through this domain and returns in Domains 2 and 16: **Meridian Care Records**, a fictional public-health programme rolling a shared clinical-records system out to **40 clinics**. Its numbers are used in KA 1.3 to make the outputs/outcomes/benefits distinction arithmetic rather than rhetorical.

KNOWLEDGE AREA 1.1

Projects, programmes and portfolios

Topics: 1.1.1 the delivery landscape · 1.1.2 project versus operational leadership · 1.1.3 the temporary organisation.

1.1.1 The delivery landscape

Definitions. A **project** is a temporary endeavour undertaken to produce a defined result. A **programme** is a group of related projects and change activities managed together to achieve **outcomes** no single project could deliver alone. A **portfolio** is the set of projects and programmes an organisation chooses to fund, selected and balanced against strategy and capacity. The three are not sizes; they are different objects of management:

Level	Optimises	Primary success test	This book
Project	Delivery of a defined output to time, cost, quality	Was the thing delivered, fit for purpose?	Domains 4–13
Programme	Coherent outcomes and benefits across components	Did the change actually land?	Domain 15
Portfolio	Value and balance of the whole investment set	Are we doing the right things at all?	Domains 2, 15

The practical consequence is that **the same event can be a project success and a programme failure**: Meridian delivering all 40 clinic installations on time is a project triumph and worth nothing if clinicians do not use the system. KA 1.3 makes that arithmetic.

1.1.2 Project versus operational leadership

Operations optimise a repeating process; projects deliver a novel result once. The differences that change a leader's behaviour:

	Operational leadership	Project leadership
Work	Repeating, refined over time	Novel, done once
Team	Stable, known, yours	Assembled, borrowed, temporary
Authority	Usually line authority	Often influence without authority
Time	Continuous	Bounded, with an end that must be planned for
Learning curve	Amortised over many cycles	Paid once, mid-delivery
Failure mode	Drift, inefficiency	Late discovery, then irreversibility

The last row is the one that shapes everything downstream. Operational problems are usually recoverable at similar cost whenever they are found; project problems get monotonically more expensive as commitments harden. That is why this book spends so much on early definition (Domain 5), honest schedules (Domain 6) and early warning (Domains 7–8) — **in temporary work, lead time is the cheapest resource a leader has, and it is spent whether used or not.**

1.1.3 The temporary organisation

A project team is a temporary organisation: people arrive with divided loyalties, no shared history, and no accumulated trust to draw on. Three consequences a leader must design for. **Trust has no time to accrue naturally**, so it must be built deliberately and early — the team charter (Domain 12) exists because the informal norms that operations grow over years must here be stated in week one. **Clarity substitutes for familiarity**: roles, decision rights and done-definitions have to be explicit precisely because nobody can infer them (Domains 3-5). **Endings must be engineered**: a temporary organisation dissolves, and the knowledge, the relationships and the operational readiness either transfer deliberately or evaporate (Domain 16).

AI in this domain — first statement of the principle

AI now sits inside the delivery environment itself: drafting plans, summarising status, generating risk lists, answering questions the team used to bring to a senior colleague. The suite's governing principle, stated once here and applied in every later domain:

AI proposes; the professional verifies, decides and remains accountable.

Three things it means concretely for a leader, before any technique in this book is applied. **Verification is a duty, not a preference** — machine output entering a plan, report or decision is checked by a named human against evidence. **Accountability is not transferable** — "the model said so" is never a defence, because accountability cannot be delegated to software (KA 1.2 makes the reason precise). **Disclosure is part of honesty** — material AI use in a deliverable is stated, so colleagues can weigh it. Domain 14 builds the full governance architecture; this domain establishes that the obligation attaches to the person, not the tool.

■ KEY TERMS — KA 1.1

Term	Meaning
Project	Temporary endeavour producing a defined result.
Programme	Related projects managed together for outcomes no one project delivers.
Portfolio	The funded, balanced set of projects and programmes.
Temporary organisation	A team with no accumulated trust or shared norms; both must be designed.
Irreversibility	The rising cost of correction as commitments harden — projects' defining failure mode.
Responsible-AI principle	AI proposes; the professional verifies, decides and remains accountable.

■ SAMPLE MCQS — KA 1.1

MCQ 1.1-A [1.1.1 · Analysis] A programme's four projects each deliver on time and to budget, but the intended service change does not materialise. The most accurate statement is: - A. the programme succeeded, since all its projects succeeded - B. the projects succeeded on their own test while the programme failed on its own test — outcomes are not the sum of outputs - C. the projects must therefore have failed - D. programme success cannot be assessed until every project closes

Rationale: Projects are judged on delivered outputs, programmes on outcomes; the two tests are different, which is exactly why the distinction exists. C rewrites history to preserve a single verdict, and D denies the measurement programmes exist to make.

MCQ 1.1-B [1.1.2 · Analysis] Why does project leadership place more weight on early definition than operational leadership does? - A. project teams are less capable - B. project problems grow monotonically more expensive as commitments harden, so lead time is the cheapest resource available - C. operations have no need for planning - D. projects have larger budgets

Rationale: Irreversibility is the structural difference (1.1.2). A and D are unfounded; C misstates operational practice, which plans continuously — just with the ability to correct cheaply next cycle.

MCQ 1.1-C [1.1.3 · Recall] A team charter is written in week one primarily because: - A. governance requires the document - B. a temporary organisation has no accumulated norms, so what operations grow informally must here be made explicit - C. it replaces the need for a project plan - D. it fixes the team's membership for the duration

Rationale: Explicit clarity substitutes for familiarity the team has had no time to build. A is a by-product, C confuses artefacts, D is false — membership usually changes.

■ SELF-CHECK — KA 1.1

1. Can a project succeed while its programme fails? — Yes: outputs delivered, outcomes not realised. The tests differ (1.1.1).
2. Name the defining failure mode of temporary work. — Late discovery followed by irreversibility; correction cost rises as commitments harden.
3. State the responsible-AI principle. — AI proposes; the professional verifies, decides and remains accountable.

KNOWLEDGE AREA 1.2

The project leader's accountability

Topics: 1.2.1 accountability and responsibility · 1.2.2 the obligation set · 1.2.3 the professional standard of care.

1.2.1 Accountability and responsibility

Definitions. Responsibility is the obligation to do the work — it can be shared and delegated. **Accountability** is the obligation to answer for the outcome — it sits with exactly one named person and **cannot be delegated**, only reassigned by whoever holds the authority to do so. A leader may hand a task to a specialist, an outsourcer or an AI tool; the answer for the result stays where it was.

Two working rules follow.

Delegation transfers responsibility, never accountability. The leader who delegates keeps the duty to specify, to verify at a proportionate depth, and to notice when the work is failing. "I gave it to the vendor" is a description of responsibility, not a defence of accountability.

One accountable person per outcome. Where two people are accountable for the same thing, nobody is: each defers, and the gap is discovered late. This is why Domain 3 builds explicit decision rights and why RACI-style matrices insist on a single "A" per row.

The AI corollary is now immediate rather than novel. A tool has no standing to be accountable — it cannot be asked to answer, cannot be sanctioned, cannot hold a duty. So every use of AI in delivery leaves accountability exactly where it was: with the professional who used the output. That is not a policy choice PCI made; it follows from what accountability is.

1.2.2 The obligation set

A project leader owes duties in four directions, and the skill is holding them together honestly rather than pretending they never conflict:

To	Owes	Typical conflict
Sponsor / organisation	Honest forecasts, stewardship of funds, no surprises	Pressure to report reassurance rather than status
Team	Clarity, safety, achievable asks, fair credit	Absorbing schedule pressure by quietly overloading people
Users / customers	A result that is genuinely fit for purpose	Delivering on time by descoping what users needed most
Public / third parties	Safety, legality, honest claims, environmental care	Cost pressure meeting a safety or compliance margin

The escalation duty. When these conflict beyond the leader's authority to resolve, the professional act is to escalate with options — not to absorb the conflict silently and hope. An unescalated conflict becomes a late surprise, and late surprises are the failure mode of §1.1.2.

The honesty asymmetry. Bad news travels badly: it costs the messenger and helps the project. A leader's most consequential cultural act is making early bad news safe — because a schedule or cost problem reported five weeks early is usually recoverable (Domain 6's case study turns on exactly this) and the same problem reported at the deadline is not.

1.2.3 The professional standard of care

Professionals are judged not on outcomes alone but on whether they exercised the **care a competent practitioner would** in the circumstances. Practically, four things constitute care in this discipline: **method appropriate to the stakes** (a two-week internal task and a safety-critical programme do not warrant the same rigour); **evidence proportionate to the claim** (a forecast presented to a board carries its basis — Domain 7's estimate basis sheet); **records that let others check the work** (decision logs, assumption registers, audit trails); and **candour about uncertainty** (ranges, not false single numbers). A leader who did all four and still missed can defend the work. One who hit the date by luck and cannot show any of them has no defence available when the next project does not get lucky.

■ KEY TERMS — KA 1.2

Term	Meaning
Responsibility	The obligation to do; shareable and delegable.
Accountability	The obligation to answer; single-holder, non-delegable.
Escalation duty	The obligation to raise, with options, a conflict beyond one's authority.
Honesty asymmetry	Bad news costs the messenger and helps the project; leaders must make it safe.
Standard of care	The care a competent practitioner would exercise in the circumstances.

■ SAMPLE MCQS — KA 1.2

MCQ 1.2-A [1.2.1 · Application] A leader outsources the migration design to a specialist vendor under a fixed-price contract. Accountability for the migration's success now rests with: - A. the vendor, under the contract - B. the leader, who retains the duty to specify, verify proportionately and act on failure - C. jointly and equally with the vendor - D. the sponsor, who approved the contract

Rationale: The contract transfers responsibility (and commercial risk); accountability is non-delegable. C describes the arrangement that guarantees nobody answers (1.2.1); D confuses approving a route with owning the delivery.

MCQ 1.2-B [1.2.1 · Analysis] An AI planning tool generates a schedule that the leader submits unchanged; it contains an infeasible dependency that causes a two-month slip. The accountability position is: - A. shared with the tool's vendor, who supplied a defective product - B. the leader's: a tool cannot answer for an outcome, and submitting unverified output is a failure of the verification duty - C. nobody's — the failure was technological - D. the team's, for not catching it

Rationale: Accountability requires a party who can be asked to answer, so it cannot attach to software (1.2.1); the specific lapse is verification (1.1's principle). Vendor recourse (A) is a commercial question and does not relocate the professional answer; D inverts delegation.

MCQ 1.2-C [1.2.3 · Analysis] A leader's project overran despite a documented method, ranged forecasts, a decision log and escalated risks. Under the standard of care this is: - A. professional negligence, because the project overran - B. defensible practice — care is judged on the exercise of competent method, not on outcome alone - C. irrelevant, since standards of care apply only to regulated professions - D. defensible only if the overrun was under 10 %

Rationale: The standard is conduct-based (1.2.3). A collapses care into outcome; C is false in substance — the expectation attaches to professional practice generally; D invents a threshold.

■ SELF-CHECK — KA 1.2

1. What can be delegated and what cannot? — Responsibility can; accountability cannot.
2. Why can accountability never attach to an AI tool? — Accountability is the obligation to answer; a tool cannot be asked to answer, sanctioned, or hold a duty.
3. Name the four constituents of care in this discipline. — Proportionate method, evidence matching the claim, checkable records, candour about uncertainty.

KNOWLEDGE AREA 1.3**Systems thinking and value**

Topics: 1.3.1 projects as systems · 1.3.2 outputs, outcomes, benefits, value · 1.3.3 leading under uncertainty.

1.3.1 Projects as systems

A project is a system: parts that interact, with **feedback** and **delay** between cause and visible effect. Three systems behaviours a leader must recognise, because each defeats common-sense management.

Delay hides consequence. Adding people to a late project consumes the productivity of those already on it (onboarding, communication paths) before adding any, so the intervention looks harmful before it looks helpful — and leaders often reverse it in the interval.

Local optimisation degrades the whole. A discipline lead maximising their own efficiency levels their team's workload smoothly and delivers the interface documents late, stalling three other teams. Every function can be efficient while the project is slow — the constraint logic of Domain 6 (6.A.1) is the schedule expression of this.

Pressure moves, it does not vanish. Squeeze the schedule and the pressure emerges as quality defects (Domain 9), scope disputes (Domain 5), team attrition (Domain 12) or supplier claims (Domain 10). A leader's question on receiving any "we absorbed it" report is where did it go?

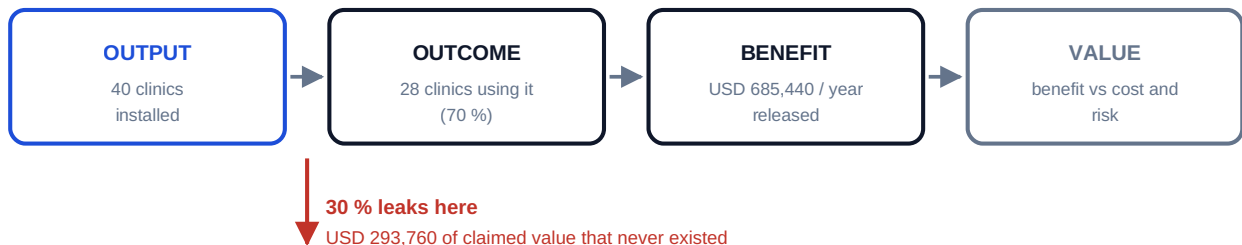
1.3.2 Outputs, outcomes, benefits and value

Definitions. An **output** is what the project delivers (a system installed, a road built). An **outcome** is the change in how things work as a result (clinicians actually using the records; journeys actually faster). A **benefit** is the measurable improvement that the outcome produces (hours released, cost avoided, harm reduced). **Value** is benefit weighed against cost and risk, for a stated stakeholder. The chain has a hard property: **each link can fail independently**, so an output claimed as a benefit is an overstatement by construction.

WORKED EXAMPLE**Worked example 1.3.2 — what Meridian actually delivers.**

1. **Setup.** Meridian Care Records installs the shared records system in **40 clinics** (the output). Realistic adoption at 12 months is **70 %** of clinics using it in daily practice. In an adopting clinic the system releases **6 clinician-hours per week**, valued at **USD 85 per hour**, over a **48-week** operating year.
2. **Formula.** Annual benefit = clinics × adoption × hours/week × rate × weeks. Contrast with the output-based claim, which counts all 40 clinics.
3. **Substitution.** Adopting clinics $40 \times 0.70 = 28$; benefit $28 \times 6 \times 85 \times 48$. Output-based claim $40 \times 6 \times 85 \times 48$.
4. **Result.** Benefit **USD 685,440 per year** ($\approx$ SAR 2.57 million indicatively). The output-based claim gives **USD 979,200** — an overstatement of **USD 293,760**, exactly **30.0 %**, which is the non-adoption rate reappearing as fictitious value.
5. **Interpretation.** The 30 % gap is not an estimating error; it is the outcome link being skipped. This is why a benefits register (Domain 16) tracks adoption as a measure in its own right, and why "40 clinics live" is a milestone, not a benefit. Sensitivity is the leader's real lever: at 50 % adoption the benefit is USD 489,600; at 90 %, USD 881,280. **A programme of this shape creates more value by moving adoption than by moving its installation date** — which should change where the leader spends personal attention (Domain 11's engagement work, not Domain 6's schedule compression).

Each link can fail independently — an output claimed as a benefit overstates by construction



Adoption is the outcome measure that links an output to any benefit at all — so it needs its own owner (Toolkit 1.T.1).

Fig 1.3.1 — The value chain, and where it leaks

1.3.3 Leading under uncertainty

Uncertainty is the condition, not an exception to be eliminated before acting. Three disciplines constitute leading well inside it.

Decide at the right moment. Deciding early buys momentum and forecloses options; deciding late preserves options and costs time. The professional question is not "do we have enough information?" but what is the last responsible moment, and what does waiting cost?

Price the time. In benefit-generating work, delay has an arithmetic cost.

WORKED EXAMPLE

Worked example 1.3.3 — what a week of Meridian delay costs.

1. **Setup.** Meridian's benefit runs at the KA 1.3.2 rate once adopted. The team can compress the rollout by **8 weeks** at an incremental cost of **USD 60,000**. Should the leader buy it?
2. **Formula.** Benefit per week = adopting clinics × hours × rate. Net = (weeks × benefit per week) – cost. Breakeven weeks = cost ÷ benefit per week.
3. **Substitution.** Per week $28 \times 6 \times 85 = 14,280$; eight weeks **114,240**; net **114,240 – 60,000**.
4. **Result.** Benefit per week **USD 14,280**; eight weeks **USD 114,240**; **net +USD 54,240**. Breakeven at **4.20 weeks** — beyond which acceleration pays.
5. **Interpretation.** The decision is not "is 60,000 a lot?" but "does it buy more than it costs?" — and here it clearly does, provided adoption materialises. Note the dependency the arithmetic makes visible: every figure rests on the 70 % adoption of 1.3.2, so **spending 60,000 to arrive sooner at an unadopted system buys nothing**. Sequencing engagement before acceleration is not a preference; it is what the numbers say.

Act reversibly where you can. Prefer decisions that can be unwound, pilots over commitments, and options kept open at low cost — then commit hard once uncertainty has genuinely reduced. This is the leadership behaviour behind rolling-wave planning (Domain 6, KA 6.3.3) and staged investment (Domain 2).

AI in this KA

Systems and value reasoning is where AI assistance is weakest and most confidently wrong: a model will produce a fluent benefits case that silently treats outputs as benefits, because that is what most documents in its training data do. The governed use is narrow and real — stress-test a benefits chain by asking what would have to be true, generate adoption scenarios, surface second-order effects a team has not considered — followed by human judgment on which link is actually fragile. The check that catches the common failure is one question: for each claimed benefit, which measured outcome produces it, and who owns that measure? (Domain 16 turns that into the benefits register.)

■ KEY TERMS — KA 1.3

Term	Meaning
Output / outcome / benefit / value	Delivered thing · change in how things work · measurable improvement · benefit against cost and risk.
Adoption	The outcome measure linking an output to any benefit at all.
Feedback and delay	Systems properties that make interventions look wrong before they look right.
Local optimisation	Function-level efficiency that degrades whole-system performance.
Last responsible moment	The latest point a decision can be taken without foreclosing a needed option.
Cost of delay	The benefit forgone per period of lateness.

■ SAMPLE MCQS — KA 1.3

MCQ 1.3-A [1.3.2 · Application] 40 clinics installed; 70 % adoption; 6 hours/week released per adopting clinic at USD 85/hour over 48 weeks. The defensible annual benefit is: - A. USD 979,200 - B. USD 685,440 - C. USD 293,760 - D. USD 14,280

Rationale: $40 \times 0.70 \times 6 \times 85 \times 48 = 685,440$. A is the output-based claim that skips adoption; C is the overstatement itself; D is the weekly benefit.

MCQ 1.3-B [1.3.2 · Analysis] A programme board is told "40 clinics live — benefits delivered". The soundest challenge is: - A. ask for the installation evidence - B. ask which measured outcome produces the benefit and who owns that measure, since installation is an output and adoption is unverified - C. accept it, since the output target was met - D. ask for the cost variance instead

Rationale: The claim skips the outcome link, which is precisely where 30 % of the value leaked in 1.3.2. A verifies the wrong thing, C repeats the error, D changes the subject.

MCQ 1.3-C [1.3.3 · Application] Benefit accrues at USD 14,280 per week once adopted. Compressing the rollout by 8 weeks costs USD 60,000. The decision and its breakeven are: - A. reject — 60,000 exceeds the weekly benefit - B. accept — net +USD 54,240, breakeven at 4.20 weeks - C. accept — net +USD 114,240 - D. indifferent — the two are equal

Rationale: $8 \times 14,280 = 114,240$, less the 60,000 cost, nets **+54,240**; breakeven $60,000/14,280 = 4.20$ weeks. A compares a total against a weekly rate; C forgets to deduct the cost; D asserts an equality the arithmetic denies.

MCQ 1.3-D [1.3.1 · Analysis] A discipline lead reports having "absorbed" a two-week schedule squeeze with no impact. The systems-literate response is: - A. record the recovery and move on - B. ask where the pressure went — into quality, scope, people or suppliers — because pressure moves rather than vanishing - C. squeeze the remaining teams equally - D. treat it as evidence the original estimate was padded

Rationale: Pressure relocates (1.3.1); the leader's job is to find where. A accepts an unverified claim, C compounds it, D leaps to a conclusion the report does not support.

■ SELF-CHECK — KA 1.3

1. Why is an output claimed as a benefit an overstatement by construction? — Because the outcome link can fail independently; unadopted outputs produce no benefit (Meridian's 30 %).
2. What is the leader's question on hearing "we absorbed it"? — Where did the pressure go?
3. What must be true before paying to accelerate Meridian? — That adoption materialises; arriving sooner at an unadopted system buys nothing.

KNOWLEDGE AREA 1.4

Ethics and the responsible use of AI

Topics: 1.4.1 professional ethics in delivery · 1.4.2 the responsible-AI principle applied · 1.4.3 the leader's AI accountability.

1.4.1 Professional ethics in delivery

Ethics in this discipline is rarely dramatic; it is a series of small reporting and allocation decisions taken under pressure. Four recurring tests:

- **Report what is, not what is wanted.** Presenting an optimistic case as the base case is a misrepresentation whatever the spreadsheet says (the same rule PFL-AI applies to forecasts).
- **Do not spend other people's margins silently.** Absorbing pressure into team overtime, a supplier's contingency or a safety margin is a decision someone is entitled to be told about.
- **Claim only what is true.** Milestone and benefit claims are statements of fact to people relying on them (1.3.2).
- **Compete and procure honestly.** Conflicts declared, evaluations run as published, suppliers not used as free consultancies (Domain 10's ethical sourcing).

Conflicts of interest are handled by disclosure and separation: declared before engagement, managed or declined — and tested by daylight, would every party, seeing the full picture, still regard this as impartial? Where the answer wavers, impartiality has already failed.

1.4.2 The responsible-AI principle applied

The principle from KA 1.1 becomes operational through four obligations that recur in every later domain:

Obligation	What it requires of a leader
Appropriate use	Match the tool to the task; prohibit it where the stakes or data forbid (Domain 14)
Verification	Machine output checked against evidence by a named human before reliance
Transparency	Material AI use disclosed in deliverables and decisions
Human decision	The judgment, and its record, belongs to a person

The failure modes to expect, stated honestly: **hallucination** (fluent, false specifics — the invented precedent, the plausible citation); **silent staleness** (confident answers from out-of-date data); **bias** inherited from training data, especially in anything touching people; **confidentiality leakage** (project data entering a tool is a disclosure); and **over-trust through fluency** — the deepest one, because polished output suppresses the scrutiny that rough output invites. None of these is a reason to refuse the technology; each is a reason the verification duty is not optional.

1.4.3 The leader's AI accountability

Three concrete leadership acts, which Domain 14 then systematises:

Name the owner. Every AI-assisted artefact that informs a decision has a human owner who verified it. Anonymous machine output has no place in a governed decision record (Domain 3, KA 3.3.4).

Make verification proportionate and real. A summary for internal reading needs a glance; a forecast in a board paper needs recomputation and a source trace; a safety-relevant analysis needs independent review. Uniform verification is either wasteful or negligent — usually both, in different places.

Protect the team's judgment. The subtler risk is not a wrong output but the atrophy of the skills needed to notice one. A leader who accepts AI-drafted plans no one on the team can critique has traded capability for speed, and will discover the price during the first recovery (Domain 6, KA 6.4).

■ KEY TERMS — KA 1.4

Term	Meaning
Daylight test	Would every party, seeing the full picture, still regard this as impartial?
Hallucination	Fluent, confidently-stated content that is false.
Over-trust through fluency	Polished output suppressing the scrutiny rough output would attract.
Verification proportionality	Depth of checking matched to the stakes of reliance.
Named owner	The human who verified an AI-assisted artefact and answers for it.

■ SAMPLE MCQS — KA 1.4

MCQ 1.4-A [1.4.1 · Analysis] A sponsor asks that the optimistic scenario be presented as the base case "to keep the board confident". The professional response is: - A. comply — scenario labels are presentational - B. decline, and offer the honest base case with sensitivities and the evidence that supports confidence - C. comply while noting the true base case privately - D. present both without indicating which is the base case

Rationale: Candour about status is a duty, not a style (1.4.1). C documents the misrepresentation without preventing it; D abdicates the professional judgment the board is relying on.

MCQ 1.4-B [1.4.2 · Application] Which AI failure mode most directly explains an authoritative-sounding schedule that encodes assumptions nobody made? - A. confidentiality leakage - B. over-trust through fluency, compounded by absent verification - C. model bias - D. hardware limitation

Rationale: Polished output invites less scrutiny (1.4.2), and the missing control is verification. A concerns data egress, C concerns systematic skew (real, but not this symptom), D is irrelevant.

MCQ 1.4-C [1.4.3 · Analysis] A leader requires the same verification depth for every AI-assisted artefact, from internal summaries to safety analyses. This is: - A. best practice — consistency is the point - B. simultaneously wasteful and negligent: proportionality means depth matched to the stakes of reliance - C. acceptable only if the team is small - D. unnecessary, since approved tools need no verification

Rationale: Uniform depth over-checks the trivial and under-checks the critical (1.4.3). A mistakes uniformity for rigour; D abandons the duty entirely.

■ SELF-CHECK — KA 1.4

1. State the daylight test. — Would every party, seeing the full picture, still regard this as impartial?
2. Which AI failure mode is hardest to counter, and why? — Over-trust through fluency: polish suppresses the scrutiny that rough work attracts.
3. What does "protect the team's judgment" mean in practice? — Do not accept AI output the team cannot critique; capability traded for speed is repaid during recovery.

— Advanced topics — Domain 1

1.A.1 Authority, influence and the borrowed team

Most project leaders hold less authority than accountability — the defining asymmetry of the role. The sources of influence that actually work in temporary organisations: **competence** (visible command of the work), **reciprocity** (having helped others deliver), **information** (being the person with a true picture), **coalition** (sponsor and peer support secured before it is needed), and **fairness** (a record of allocating credit and blame accurately). Positional authority is the weakest of the six in borrowed teams, because the borrowed people's careers are decided elsewhere. The practical implication: influence is built in advance, and a leader who begins building it during a crisis is too late.

1.A.2 Success criteria and the multiple-verdict problem

Projects are judged simultaneously against delivery (time, cost, quality), outcome and benefit realisation, and stakeholder satisfaction — which is why the same project is often described as a success and a failure by different parties, honestly. The professional response is not to pick a flattering verdict but to **agree the criteria in advance, with owners and measures** (Domain 2's benefits mapping, Domain 5's acceptance criteria), and to report against all of them. Unagreed success criteria are settled retrospectively by whoever is most disappointed.

1.A.3 The reviewer's leadership eye

Invariants an experienced reviewer tests in the first hour on any project: exactly one accountable name per outcome; a benefits chain where each claimed benefit traces to a measured outcome with an owner; decision records that show what was known when; escalations that arrived early enough to matter; forecasts with ranges rather than single numbers; and AI-assisted artefacts carrying a named verifier. Every one of these is cheap to check and expensive to have missing — the same posture Domains 6 and 7 apply to schedules and cost.

— Industry variations — Domain 1

- **Public programmes.** Accountability is doubled — professional and political — and the public obligation of 1.2.2 is enforceable through scrutiny bodies; benefits (not outputs) are the currency of accountability, and announced dates often precede plans (Domain 6's Case B).
- **Regulated industries (pharma, aviation, nuclear).** The standard of care is codified and auditable; the "care exercised" defence of 1.2.3 is a documented reality rather than an argument, and AI use faces explicit validation expectations.
- **Construction and engineering.** Authority is contractual as much as organisational; influence runs through commercial relationships, and the leader's obligation set includes site safety as a first-order duty.
- **Technology and product.** Outcome ownership frequently sits with a product role rather than the delivery leader, making the 1.3.2 chain a shared accountability that must be explicitly split — the commonest source of "who owns adoption?" disputes.
- **Professional services and consulting.** Duties to the client organisation and to the end-users can diverge, and independence (1.4.1) becomes a standing rather than occasional test.

— Case study — Domain 1: Meridian under scrutiny (public health)

Situation. Eighteen months in, Meridian Care Records reports 40 of 40 clinics installed, on time, 3 % under budget. The programme is publicly praised. Nine months later a health-service audit finds clinician hours released are running at roughly a third of the business case, and the programme is publicly described as a failure. Both accounts cite true facts.

Analysis. The delivery verdict was sound: the outputs landed. The benefits case had been written against **outputs** — all 40 clinics, no adoption term — so it claimed USD 979,200 per year where the honest figure at 70 % adoption was USD 685,440, and actual adoption reached only about 40 %, giving roughly USD 391,680. Two failures compounded: an arithmetic one (skipping the outcome link, 1.3.2) and an accountability one (**nobody owned adoption** — the delivery leader owned installations, the clinical directorate owned practice change, and no single name owned the measure that connected them, 1.2.1).

What was done. The recovery was not technical. Adoption was made a named accountability with a monthly measure; the benefits case was restated with an adoption term and a sensitivity range (50/70/90 % → 489,600 / 685,440 / 881,280); clinical champions were funded in the eleven lowest-adoption clinics; and the programme board's report changed from "clinics live" to "clinics using, and hours released". Adoption reached 68 % within a year.

What the domain teaches here. Every element of this domain appears: outputs are not benefits (1.3.2); one accountable name per outcome (1.2.1); success criteria agreed in advance or settled by the disappointed (1.A.2); and the leader's attention belongs where the value actually leaks — which the arithmetic, not instinct, identified.

— Case study B — Domain 1: the plan nobody could critique (financial services)

Situation. A platform-migration leader under time pressure used an AI planning assistant to generate the full delivery schedule and dependency network, reviewed it briefly, and issued it. It was detailed, internally consistent and professionally formatted. It also assumed the legacy data extract could run concurrently with the schema freeze — an assumption no one on the team had made, and which the two engineers who would have spotted it never saw.

What happened. The conflict surfaced eleven weeks later during rehearsal, costing a seven-week slip and a weekend of reputational damage with the regulator's supervisory team. The post-incident review found no fault in the tool: the schedule was a reasonable inference from an under-specified prompt. The failures were the leader's — no verification against the discipline leads (the 1.1 principle and 1.4.2 obligation), and no named verifier on the artefact (1.4.3). The compounding finding was cultural: the polish of the output had actively discouraged challenge, and two team members later said they had assumed the network "must have been checked".

What the domain teaches here. Fluency is not evidence. Accountability did not move to the vendor or the model — it stayed with the person who issued the plan (MCQ 1.2-B is this case in miniature). And the durable fix is structural, not exhortative: a named verifier per artefact, discipline leads walking their own links, and a team kept capable of critique.

— Executive perspective — Domain 1

What a project leader cannot delegate in this domain:

- **The single accountable name.** For every outcome, including the awkward ones like adoption. Where two names appear, the leader's job is to reduce it to one.
- **The benefits chain's honesty.** Each claimed benefit traced to a measured outcome with an owner. A leader who lets outputs be reported as benefits has authorised the failure of Meridian's first eighteen months.
- **The obligation set, held openly.** Sponsor, team, users, public — with conflicts escalated with options rather than absorbed in silence.
- **Bad news made safe.** The single highest-leverage cultural act available, because it converts irreversible surprises into recoverable problems.
- **The AI accountability line.** Named verifiers, proportionate depth, and a team that remains able to critique what the tools produce.

— Calculation exercises — Domain 1

Exercise 1.1 A programme installs a system in 60 sites; adoption is 65 %; each adopting site releases 5 hours/week at USD 90/hour over 46 weeks. Compute the defensible annual benefit and the overstatement in an output-based claim. Solution. Adopting sites $60 \times 0.65 = 39$; benefit $39 \times 5 \times 90 \times 46 = \text{USD } 807,300$. Output-based $60 \times 5 \times 90 \times 46 = \text{USD } 1,242,000$; overstatement $\text{USD } 434,700 = 35.0 \%$, exactly the non-adoption rate. Common error: applying adoption to the rate rather than the site count — the same product here, but it conceals which link failed and misdirects the fix.

Exercise 1.2 Using Meridian's figures (28 adopting clinics, 6 h/week, USD 85/h), compute the cost of a 5-week delay and the maximum justified acceleration spend for 5 weeks. Solution. Per week $28 \times 6 \times 85 = \text{USD } 14,280$; five weeks $\text{USD } 71,400$ — which is also the maximum justified spend, since above it acceleration destroys value. Common error: comparing the spend against annual benefit rather than the benefit for the weeks actually saved.

Exercise 1.3 Meridian's benefit at 50 %, 70 % and 90 % adoption (40 clinics, 6 h/week, USD 85/h, 48 weeks). What does the spread imply for where the leader spends attention? Solution. $\text{USD } 489,600 \cdot 685,440 \cdot 881,280$. The 40-point adoption swing moves annual benefit by $\text{USD } 391,680$ — far more than plausible schedule compression could add (Exercise 1.2 values 5 weeks at 71,400). Attention belongs on adoption (Domain 11), not acceleration.

— Practitioner's toolkit — Domain 1

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 1.T.1 — Leader's accountability map (one page)

Per outcome: the outcome stated in outcome language (not output language) · **the single accountable name** · responsible parties · the measure that evidences it and its owner · the decision

authority and threshold (Domain 3) · review cadence. Rule: if any row has two accountable names or an output in the outcome column, it is not finished.

Toolkit 1.T.2 — Benefits-chain test (run on any benefits claim)

- [] Each claimed benefit names the measured **outcome** that produces it.
- [] Each outcome measure has an owner and a reporting cadence.
- [] Adoption (or the equivalent behavioural change) appears explicitly, with a value.
- [] The claim carries a sensitivity range, not a single number.
- [] Output milestones are labelled as milestones, never as benefits.
- [] Cost of delay per period is stated, so timing decisions can be priced (1.3.3).

Toolkit 1.T.3 — AI-use and verification record

Per AI-assisted artefact: tool and approved environment · data classification cleared · prompt/intent summary · **verification performed and its depth** · **named verifier** · disclosure status in the deliverable · date. Standing agenda item, not an archive: the register's purpose is that the next artefact is verified, not that the last one was.

— Exam preparation — Domain 1

The traps. Treating outputs as benefits (1.3.2 — the domain's central trap, and Meridian's whole case study) · saying accountability is "shared" (1.2.1) · assigning accountability to a vendor or a tool (MCQ 1.2-A, 1.2-B) · comparing an acceleration cost against annual rather than saved-period benefit (Exercise 1.2) · reading a delivery success as a programme success (1.1.1) · accepting "we absorbed it" (1.3.1) · uniform AI verification depth (1.4.3) · judging the standard of care by outcome alone (1.2.3).

Reflection questions. 1. Take your current project: write the single accountable name for its principal outcome. If you hesitated, or wrote two, what does that predict? 2. For each benefit in your business case, which measured outcome produces it and who owns that measure? What percentage of your claimed value survives the question? 3. Which AI-assisted artefact in your last month of work had no named verifier — and what would have caught it if it had been wrong?

— Domain 1 summary

Project leadership is a distinct profession because temporary work fails distinctively: problems discovered late become irreversible, teams have no accumulated trust to draw on, and lead time is the cheapest resource available and is spent whether used or not. The discipline rests on a precise view of accountability — responsibility is delegable, the obligation to answer is not, one name per outcome — held across four directions at once (sponsor, team, users, public) with conflicts escalated rather than absorbed, and judged by the care a competent practitioner would exercise rather than by outcome alone. Two habits of mind make a leader rather than a coordinator: systems thinking, which expects feedback, delay, local optimisation and pressure that relocates instead of vanishing; and the outputs-outcomes-benefits-value chain, which Meridian turns into arithmetic — 40 clinics installed produce USD 685,440 of annual benefit at 70 % adoption, not the USD 979,200 an output-based claim asserts, and the 30 % gap is the outcome link being skipped. Timing decisions are priceable (USD 14,280 per week; acceleration breaking even at 4.20 weeks), and attention belongs where value actually leaks. Around all of it sits professional ethics and the suite's governing principle — **AI**

proposes; the professional verifies, decides and remains accountable — with named verifiers, proportionate depth, and a team kept capable of critique. Domain 2 turns strategy into selected work; Domain 3 gives the accountability of this domain its governance machinery.

06

DOMAIN 6

Planning, Scheduling and Delivery Flow (quantitative flagship)

Group: Delivering the work (Domain 6 of 6 in Part Two). **Target:** ~76 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries (<docs/books/registries/>). This domain is the home of the schedule symbols — **ES**, **EF**, **LS**, **LF**, **TF** (total float), **FF** (free float) — restated by every domain that touches time. British English; USD (+SAR where useful, indicative **USD 1 ≈ SAR 3.75**).

— Why this domain exists

A project leader's authority rests on one repeated act: making a credible statement about the future — when the work will finish, what could move that date, and what it would cost to move it back. This domain builds the machinery behind that credibility. It starts where planning actually starts — with levels of plan and the logic that binds work together (KA 6.1); builds the critical path method in full, forward and backward passes, and both kinds of float (KA 6.2); adds the realities the textbook network ignores — resources, constraints, milestones and rolling-wave elaboration (KA 6.3); and finishes with delivery flow across predictive, agile and hybrid worlds: recovery, compression economics, and forecasting under uncertainty (KA 6.4). Cost joins schedule in Domain 7 (earned value); risk quantification deepens in Domain 8. A leader who cannot read a network diagram is hostage to whoever can; this domain removes the hostage-taking.

Learning objectives. After this domain a candidate can: choose the right planning level for an audience and a decision; build a logic network with correct dependency types; run forward and backward passes and derive total and free float; identify the critical path and near-critical paths and explain why float is not spare time; plan resources against the schedule and level them deliberately; apply rolling-wave elaboration honestly; select and price schedule compression; distinguish predictive, agile and hybrid scheduling and govern each; and use three-point estimates and scenario analysis to forecast completion with stated uncertainty — with any AI-generated schedule subject to the family's verification rule.

The master worked project. One project runs through this domain and returns in Domains 7 and 8: **Project Auriga**, a control-systems upgrade for a regional utility, planned in weeks:

ID	Activity	Duration (wk)	Predecessors
A	Mobilise	2	—
B	Detailed design	6	A
C	Procure control hardware	8	B
D	Civil and cabling works	7	B
E	Installation	5	C, D
F	Testing and commissioning	4	E
G	Operator training and handover	2	D

Every calculation below uses this network.

KNOWLEDGE AREA 6.1

Planning levels and logic networks

Topics: 6.1.1 levels of plan · 6.1.2 dependencies and logic · 6.1.3 network rules and quality.

6.1.1 Levels of plan

The principle. One schedule cannot serve a board, a site supervisor and a lender at once. Mature delivery organisations maintain a **hierarchy of schedules**, each derived from the one below, each honest with the one above:

Level	Name	Audience & decision	Typical granularity
L1	Executive summary	Board, sponsor — go/no-go, milestones	10–30 bars
L2	Management schedule	Steering, PMO — stage gates, interfaces	50–300 activities
L3	Control schedule	Project team — the CPM network; progress and float live here	300–5,000 activities
L4	Execution/lookahead	Site and squads — 2–6 week detail	daily/shift detail

The **control schedule (L3)** is where this domain's machinery operates: it carries the logic, the floats and the critical path, and it is the version under change control (Domain 4, KA 4.3). An L1 milestone that cannot be traced to L3 logic is decoration, and steering committees should treat it as such (Domain 3, KA 3.2).

Fig 6.1.1 — The schedule hierarchy: one logic, four honest views

6.1.2 Dependencies and logic

Definitions. Activities link through four dependency types — finish-to-start (**FS**, the default: design finishes, then procurement starts), start-to-start (**SS**: cabling can start once civils have started, often with a lag), finish-to-finish (**FF**: commissioning documentation finishes when commissioning finishes),

and the rare start-to-finish (**SF**). A **lag** delays a successor (**FS+2** = start two weeks after the predecessor finishes); a **lead** (negative lag) overlaps it. Logic expresses physics and choice — what must follow, and what the team has chosen to sequence — and the two should be distinguishable in the schedule's notes, because only chosen logic can be re-chosen when recovery demands it (KA 6.4).

Common pitfall — the lag that hides work. A 3-week lag labelled "curing" is physics; a 3-week lag labelled nothing is often an activity somebody didn't model — unresourced, uncosted, invisible to progress measurement. Audit rule: every lag has a stated reason, and long lags (> 10 % of project duration) are activities in disguise.

WORKED EXAMPLE

Worked example 6.1.2 — re-choosing a dependency.

1. **Setup.** Survey work P (4 weeks) precedes report drafting Q (6 weeks), currently linked **FS+2** (two weeks of data cleaning after all surveying ends). The team proposes drafting in parallel as survey sections complete: **SS+1**. Compute Q's dates both ways (P starts week 0).
2. **Formula.** **FS+lag:** $ES(Q) = EF(P) + \text{lag}$. **SS+lag:** $ES(Q) = ES(P) + \text{lag}$.
3. **Substitution.** **FS+2:** $ES(Q) = 4 + 2 = 6$, $EF(Q) = 12$. **SS+1:** $ES(Q) = 0 + 1 = 1$, $EF(Q) = 7$.
4. **Result.** The chain finishes week **12** under **FS+2** and week **7** under **SS+1** — five weeks earlier from one logic choice.
5. **Interpretation.** Nothing was crashed and nobody works faster: the saving comes from overlapping, and its price is rework risk if early sections change after later surveying. This is why chosen logic must be visibly chosen (not fossilised as physics) — it is the cheapest recovery currency the project owns (KA 6.4.2), and spending it is a risk decision, not a scheduling trick.

6.1.3 Network rules and quality

A network the passes can trust obeys checkable rules: every activity except the first has a predecessor and except the last a successor (**no dangles**); no loops; durations estimated by the owning discipline, not the scheduler; no date constraints doing logic's job (a "must start on" pin that overrides logic converts the schedule from a model into a poster); and logic density in a healthy range — too few links and float is fiction, too many and nothing can move. These rules are the schedule-quality gate a leader can run without scheduling software: ask for the dangle count, the constraint count and the reasons for both.

AI in this KA

Scheduling tools now draft networks from scope statements and historical libraries. Useful — the draft logic is often 80 % right — and dangerous in a specific way: generated logic looks authoritative while encoding assumptions nobody made. The governed workflow: AI proposes the network; the discipline leads walk every link of their scope ("does B truly need all of A finished?"); the scheduler runs the quality rules above; and the leader signs the logic as a decision record (Domain 3, KA 3.3.4).

AI proposes; the professional verifies, decides and remains accountable.

■ KEY TERMS — KA 6.1

Term	Meaning
Schedule hierarchy (L1-L4)	Nested plans for different audiences, derived from one logic.
Control schedule	The L3 CPM network under change control; home of float and progress.
FS / SS / FF / SF	The four dependency types.
Lag / lead	Imposed delay / overlap on a dependency.
Dangle	An activity missing a predecessor or successor; a network defect.
Date constraint	A pinned date overriding logic; each one weakens the model.

■ SAMPLE MCQS — KA 6.1

MCQ 6.1-A [6.1.2 · Application] Cabling may begin one week after civil works begin. The correct dependency is: - A. FS+1 - B. SS+1 - C. FF+1 - D. FS-1

Rationale: The condition binds the two starts with a one-week lag: start-to-start plus one. A would wait for all civils to finish; C binds the finishes; D (a lead on FS) overlaps the finish, which is not what was stated.

MCQ 6.1-B [6.1.3 · Analysis] A schedule shows 14 "must finish on" constraints, all on milestones reported to the board. The most accurate reading is: - A. the schedule is well-governed, because board dates are protected - B. the milestones will be met, because the tool will honour the constraints - C. the schedule may no longer model reality — pinned dates can mask logic-driven slippage that float analysis would otherwise reveal - D. constraints are neutral scheduling hygiene with no analytical effect

Rationale: Pins override logic: the network can be slipping while pinned milestones hold still, hiding negative float until it is unrecoverable. A and B mistake suppression for control; D is false — every pin degrades the model's predictive value.

MCQ 6.1-E [6.1.1 · Recall] The schedule level that carries the logic, the floats and the critical path — and sits under formal change control — is: - A. L1, because the board owns the schedule - B. L2, because the PMO maintains it - C. L3, the control schedule - D. L4, because it has the most detail

Rationale: The L3 control schedule is the analytical model; L1/L2 summarise it and L4 elaborates near-term execution from it. Detail (D) is not the same as control — L4 churns weekly by design and is never the baselined network.

MCQ 6.1-C [6.1.2 · Application] Survey P (4 wk) feeds report Q (6 wk). Under FS+2 the chain completes week 12. Re-linked SS+1, it completes week: - A. 7 - B. 11 - C. 12 - D. 5

Rationale: $ES(Q) = ES(P) + 1 = 1$, $EF(Q) = 7$. B subtracts one lag week from 12; C assumes logic changes nothing; D forgets Q's own duration must still run.

MCQ 6.1-D [6.1.3 · Recall] A "dangle" in a schedule network is: - A. an activity with negative float - B. an activity missing a predecessor or successor link - C. a milestone with zero duration - D. a lag longer than its predecessor

Rationale: Dangles are unbound activity ends — the passes cannot constrain them, so their dates and float are unreliable. A describes constraint-driven lateness; C describes every milestone; D is unusual but legal when justified.

■ SELF-CHECK — KA 6.1

1. Why must chosen logic be distinguishable from physical logic? — Because recovery (KA 6.4) works by re-choosing choices; physics cannot be renegotiated.
2. What three numbers give a fast schedule-quality read? — Dangle count, constraint count, unexplained-lag count — each should be at or near zero, with reasons for the rest.

KNOWLEDGE AREA 6.2

The critical path and float

Topics: 6.2.1 the forward and backward passes · 6.2.2 total and free float · 6.2.3 reading the critical path.

6.2.1 The forward and backward passes

Definitions. For each activity: **ES/EF** — earliest start/finish (forward pass, left to right: **ES** = latest **EF** of predecessors; **EF** = **ES** + duration); **LS/LF** — latest start/finish that do not delay the project (backward pass, right to left: **LF** = earliest **LS** of successors; **LS** = **LF** - duration). The project duration is the largest **EF** in the network.

Worked example 6.2.1 — Auriga, both passes. The passes are inherently tabular; the labelled table replaces the five-step skeleton, and the interpretation follows (pattern rule).

Act	Dur	ES	EF	LS	LF	TF	FF
A	2	0	2	0	2	0	0
B	6	2	8	2	8	0	0
C	8	8	16	8	16	0	0
D	7	8	15	9	16	1	0
E	5	16	21	16	21	0	0
F	4	21	25	21	25	0	0
G	2	15	17	23	25	8	8

Interpretation. Project duration **25 weeks**; the **critical path is A-B-C-E-F** ($2+6+8+5+4 = 25$), every activity on it with zero float. D can slip one week without moving the end date; G can slip eight. The passes turn a drawing into an instrument: every schedule statement a leader makes in this domain — "we finish in week 25", "D matters more than G", "procurement drives the job" — is read directly off this table.

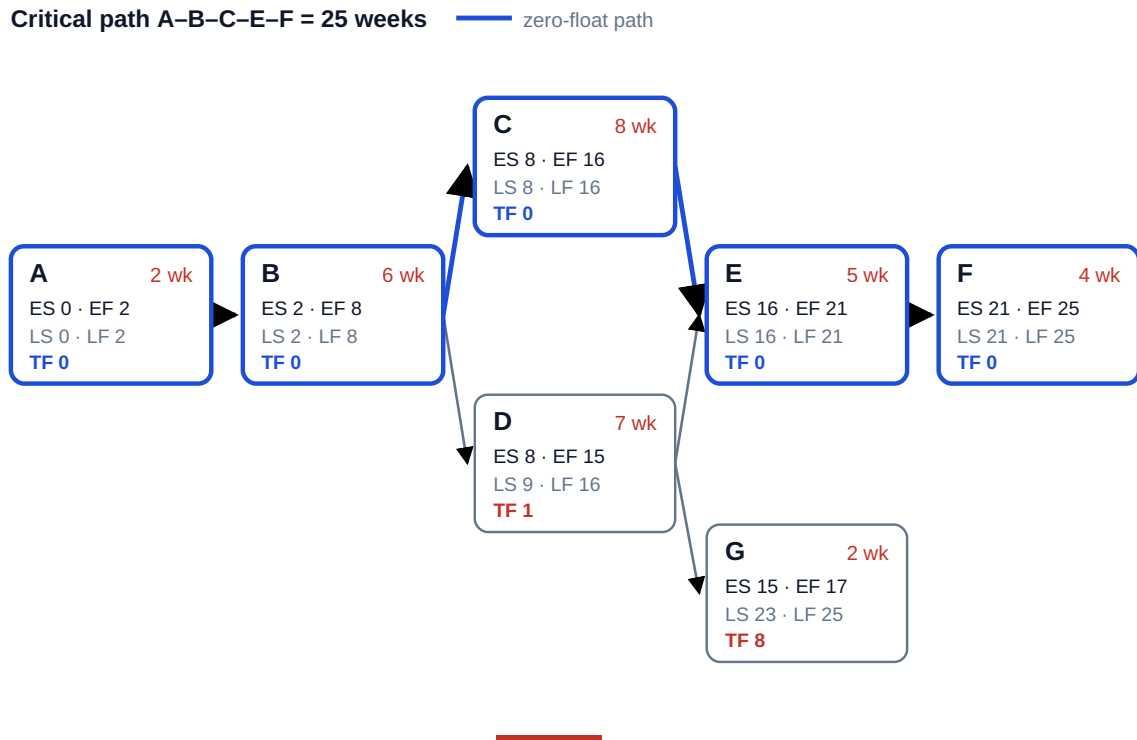


Fig 6.2.1 — The Auriga network with both passes

6.2.2 Total and free float

Definitions.

$TF = LS - ES$ — how far an activity can slip without delaying the project
 $FF = \min(ES \text{ of successors}) - EF$ — how far it can slip without delaying ANY successor

$FF \leq TF$ always. The Auriga table carries the domain's subtlest lesson: **D has $TF = 1$ but $FF = 0$** — D can slip a week without moving the end date, but the moment it slips at all, G's earliest start moves. Total float is a project-level commodity; free float is a courtesy to your successors. A leader who grants a subcontractor "the float" without saying which kind has just given away someone else's schedule.

Float is not spare time. Float belongs to the path, not the activity: consume D's week and every activity sharing that path inherits zero. Contract regimes differ on who owns float (Domain 10, KA 10.3, treats float ownership clauses); the delivery answer is simpler — float is a risk buffer under the leader's governance, spent deliberately or not at all.

Common pitfall — managing only the critical path. D sits one week from critical. A team watching only A-B-C-E-F will discover D's importance the week it becomes too late to matter. Near-critical analysis (all paths within, say, 10 % of project duration) is standing practice; KA 6.4's scenario work builds on it.

6.2.3 Reading the critical path

The critical path is the longest path and therefore the shortest possible duration — both statements are the same fact, and fluency means holding both. Three leader-grade readings of the Auriga table: **(1)** procurement (C) is the single most schedule-driving activity — an expediting conversation is worth more than any amount of site pressure; **(2)** the C/D pair converge on E, so E's start is hostage

to the later of two chains — convergence points are where slippage compounds and where progress review should focus (Domain 8, KA 8.1); **(3)** G's eight weeks of float make it the natural donor of resources in any recovery (KA 6.4).

AI in this KA

CPM arithmetic is deterministic — ideal for machine computation and instant human verification. The two checks that catch nearly every pass error: the critical path's durations must **sum exactly** to the project duration, and every critical activity must show $TF = 0$ with $ES = LS$. An AI-produced schedule that fails either is broken; one that passes both may still have wrong logic — which is why KA 6.1's link-walking discipline precedes trust in any pass, human or machine.

■ KEY TERMS — KA 6.2

Term	Meaning
Forward / backward pass	Left-to-right ES/EF computation; right-to-left LS/LF computation.
ES EF LS LF	Earliest/latest start and finish per activity.
Total float TF	$LS - ES$; slip available without delaying the project.
Free float FF	Slip available without delaying any successor; $FF \leq TF$.
Critical path	The zero-float longest path; the project's shortest possible duration.
Convergence point	A node where paths merge; slippage compounds there.

■ SAMPLE MCQS — KA 6.2

MCQ 6.2-A [6.2.1 · Application] In the Auriga network, activity D (duration 7, predecessor B finishing week 8, successors E and G) has $LS = 9$ and $ES = 8$. Its total float is: - A. 0 - B. 1 - C. 8 - D. 7

Rationale: $TF = LS - ES = 9 - 8 = 1$. A confuses D with the critical activities; C is G's float; D is the activity's duration, not its float.

MCQ 6.2-B [6.2.2 · Analysis] An activity shows $TF = 1$ and $FF = 0$. Delaying it by one week will: - A. delay the project by one week - B. delay nothing, because total float absorbs the slip - C. delay at least one successor's earliest start while leaving the project end date unchanged - D. be impossible — free float can never be below total float

Rationale: Positive TF protects the end date (not A); zero FF means some successor moves immediately (not B). D inverts the invariant — $FF \leq TF$ always.

MCQ 6.2-C [6.2.3 · Application] Auriga's procurement activity C is crashed from 8 weeks to 6. The new project duration is: - A. 23 weeks - B. 24 weeks - C. 25 weeks - D. 21 weeks

Rationale: After one week of crashing, path B-D-E ($8+7+5+4$ via D) becomes co-critical at 24; the second crashed week buys nothing — duration stays 24. A assumes both weeks convert to project weeks; C forgets the crash entirely; D subtracts from the wrong baseline. (The full economics: KA 6.4.)

MCQ 6.2-F [6.2.1 · Application] Auriga's procurement C slips from 8 weeks to 9 (all else unchanged). The new project duration is: - A. 25 weeks - B. 26 weeks - C. 27 weeks - D. 24 weeks

Rationale: C was critical with zero float, so its extra week passes straight through: E runs 17–22, F finishes week 26. A assumes float absorbed a critical activity's slip; C adds the week twice; D subtracts it.

MCQ 6.2-D [6.2.1 · Application] Auriga's training activity G runs ES 15–EF 17 with LF = 25. Its total float is: - A. 8 - B. 10 - C. 2 - D. 0

Rationale: $TF = LF - EF = 25 - 17 = 8$ (equivalently $LS - ES = 23 - 15$). B ignores G's duration (25 – 15); C is G's duration; D confuses G with the critical path.

MCQ 6.2-E [6.2.3 · Analysis] A programme's binding path shows $TF = -3$ after a constraint-honest pass. The professional reading is: - A. the software has malfunctioned — float cannot be negative - B. the plan, as constrained, is already three weeks late; the number sizes the recovery problem and must be reported now - C. delete the constraints so the float returns to zero - D. the project will finish three weeks early

Rationale: Negative float is the gap between what logic needs and what constraints allow — information, not error (A) and not good news (D). C is the pin-and-squeeze suppression this domain's Case B dissects; it hides the problem until it is unrecoverable.

■ SELF-CHECK — KA 6.2

1. State both definitions of the critical path and why they are equivalent. — Longest path through the network; shortest possible project duration — the longest chain of unavoidable work is precisely what bounds how soon the whole can finish.
2. Why is $FF \leq TF$ always? — Delaying any successor is one of the ways of delaying the project; the constraint set defining FF is stricter.
3. Auriga slips: D takes 10 weeks, not 7. New duration? — 27 weeks: D finishes week 18, E waits for it ($18 > 16$), E–F add 9 → the critical path re-routes through D (KA 6.4 works the recovery).

KNOWLEDGE AREA 6.3

Resources, constraints, milestones and rolling wave

Topics: 6.3.1 resource planning and levelling · 6.3.2 constraints and milestones · 6.3.3 rolling-wave planning.

6.3.1 Resource planning and levelling

The principle. The passes of KA 6.2 assume infinite resources; reality staffs the network. Loading resources onto the schedule produces a **histogram** — demand per period — and the histogram almost always spikes. Two responses: **resource smoothing** uses float to flatten demand without moving the end date (slip D and G inside their float to pull crews off the peak); **resource levelling** accepts a later end date if the resource cap is hard. The order of operations is a leadership decision in disguise: smoothing spends float — the project's risk buffer — to save money, and someone accountable should price that trade explicitly, not discover it in a progress meeting.

WORKED EXAMPLE**Worked example 6.3.1 — smoothing Auriga's field crews.**

1. **Setup.** Weeks 9-15: civil works D needs 3 crews; early installation staging inside C needs 2; the site cap is 4 crews. Overlap weeks 9-15 demand 5 — one over cap.
2. **Formula.** Smoothing test: can a demanding activity slip within **TF** (and, if successors matter, **FF**) to clear the peak?
3. **Substitution.** D has **TF** = 1, **FF** = 0: slipping D one week clears one peak week but consumes D's entire float and immediately moves G (**FF** = 0). The staging work inside C has discretionary timing across weeks 9-16.
4. **Result.** Re-sequence the staging within C to weeks 14-16 where D's demand tails off: peak demand falls to 4, the cap holds, the end date holds, and **no float is spent**.
5. **Interpretation.** The cheapest capacity is re-sequencing; the second cheapest is float; the expensive ones are overtime, second shifts and extension. A leader asks for the histogram with the options priced, not just the spike.

WORKED EXAMPLE**Worked example 6.3.1b — when the cap is hard: levelling, priced.**

1. **Setup.** Now suppose the site cap is **3 crews** (a permit condition, not a preference). D needs all 3 for weeks 9-15; the 4-week staging package (2 crews) must finish before E can start (week 16). Delay costs USD 45,000 per week (the client's bonus forgone); a second-shift waiver for the staging crew costs USD 20,000 per week. What does the leader do?
2. **Formula.** Compare total cost of each feasible plan: pure levelling (extend) vs paid capacity (second shift) vs any float-funded re-sequence.
3. **Substitution.** Levelling: staging cannot start until D releases crews (week 16); it runs weeks 16-19, E starts week 20 — project 25 → 29, cost $4 \times 45,000 = 180,000$. Second shift: staging runs weeks 12-15 off-shift, E starts on time — cost $4 \times 20,000 = 80,000$. No float-funded option exists: D's **TF** = 1 cannot host a 4-week move.
4. **Result.** **Buy the second shift: USD 80,000** against USD 180,000 of extension — and the end date holds.
5. **Interpretation.** A hard cap converts a scheduling problem into a procurement problem. The histogram's job is to surface that conversion early enough for the cheap option to exist — second shifts, pre-assembly, off-site staging all have lead times of their own. Levelling that silently extends the project is a decision someone should have been asked to make.

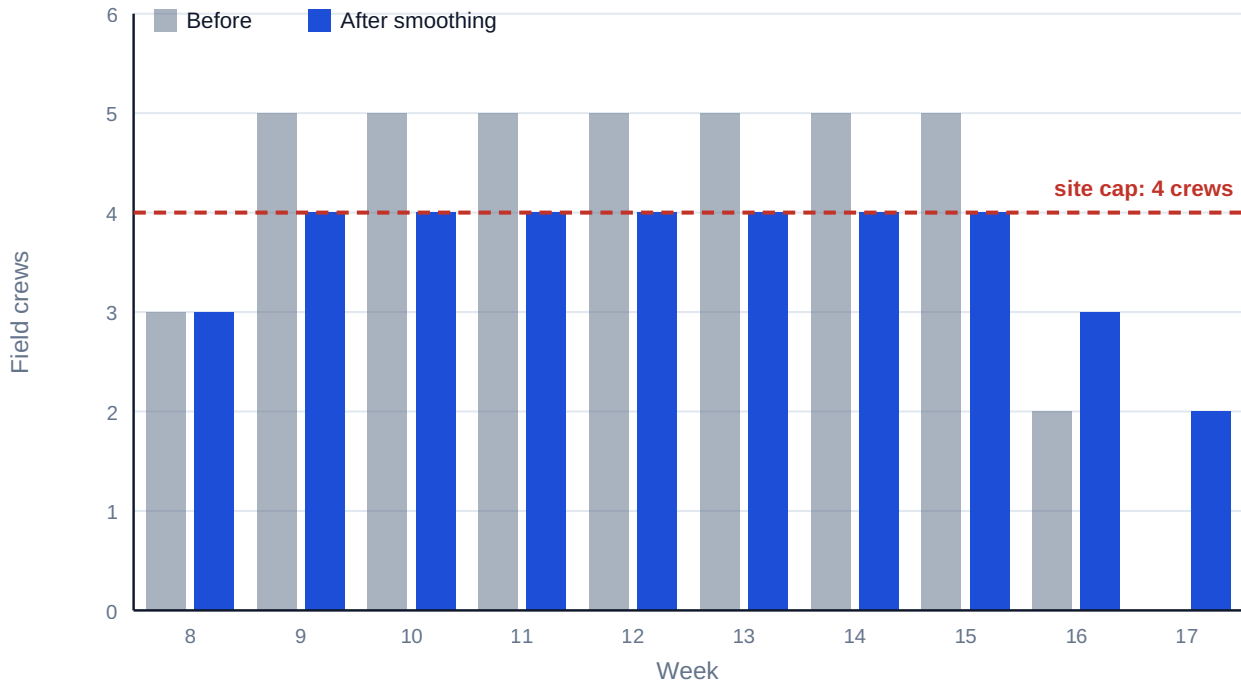


Fig 6.3.1 — Auriga field-crew histogram, before and after smoothing

6.3.2 Constraints and milestones

Constraints are externally imposed dates — an outage window, a regulatory deadline, a seasonal limit. Modelled honestly they are logic (an outage window is a predecessor with a date); modelled lazily they are pins that falsify float (KA 6.1.3). **Milestones** are zero-duration events that carry meaning: contractual obligations (Domain 10), gate decisions (Domain 3), integration points with other projects (Domain 15). A milestone's date is an output of the network, never an input — the moment a milestone is typed rather than computed, the schedule has stopped forecasting and started decorating.

Milestone hygiene: each has an owner, an unambiguous done-definition (Domain 5's acceptance criteria), traceable L3 logic, and — for contractual milestones — a stated float position reported honestly to the counterparty (Domain 11's reporting ethics).

6.3.3 Rolling-wave planning

The principle. Detail decays with distance: estimating month 18's tasks at daily granularity today manufactures precision, not knowledge. **Rolling wave** plans near-term work at execution detail (L4) and far work at planning packages (L3 summary), elaborating each wave as it approaches under change control.

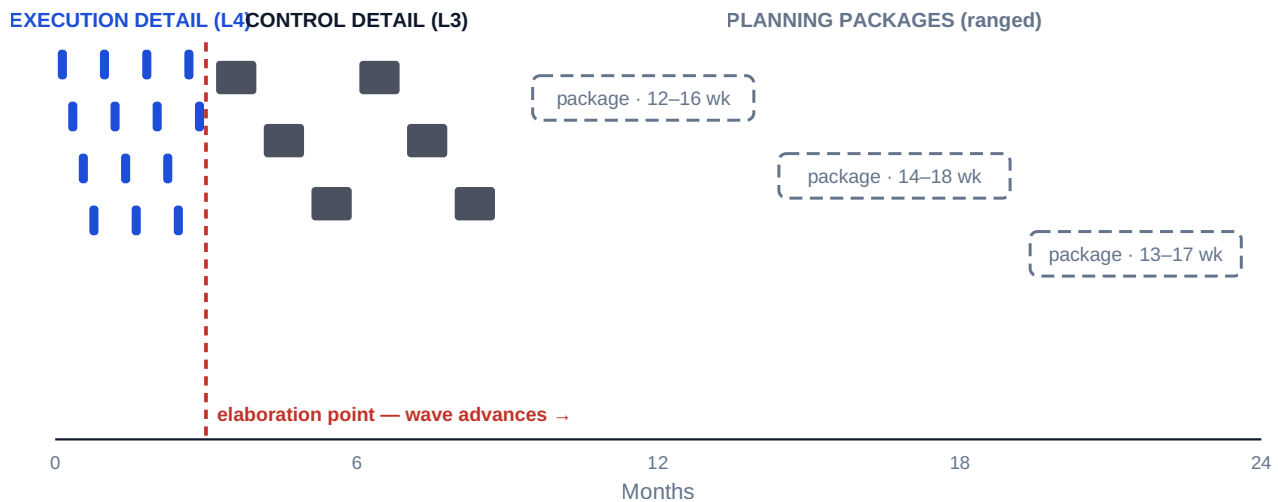


Fig 6.3.2 — The rolling-wave horizon

AI in this KA

Resource optimisation is a genuine AI strength — levelling across thousands of activities and dozens of calendars is combinatorial work machines do better than planners. Governance holds the same shape: the optimiser proposes a levelled plan; the planner verifies the constraint set was real (crew interchangeability, shift rules, site logistics); the leader owns the trade the optimiser cannot see — whether spending float, money or scope is the right currency this month. An optimiser given a wrong constraint produces a beautifully efficient fiction.

■ KEY TERMS — KA 6.3

Term	Meaning
Resource histogram	Demand per period from the loaded schedule.
Smoothing / levelling	Flattening demand within float / accepting a later date under a hard cap.
Constraint	An externally imposed date; modelled as logic, not as a pin.
Milestone	Zero-duration event with an owner and a done-definition; its date is computed.
Rolling wave	Near-term detail, far-term packages, elaborated under change control.
Planning package	A far-wave summary activity with ranged duration awaiting elaboration.

■ SAMPLE MCQS — KA 6.3

MCQ 6.3-A [6.3.1 · Application] A planner clears a resource peak by delaying an activity with $TF = 1$, $FF = 0$ by one week. The immediate consequences are: - A. no schedule effect of any kind - B. project delay of one week - C. the end date holds, the activity's path float is exhausted, and at least one successor's earliest start moves - D. the resource peak worsens

Rationale: One week inside TF protects the end date (not B), but zero FF moves a successor and the path's buffer is now spent (not A) — the smoothing was legal but not free.

MCQ 6.3-B [6.3.3 · Analysis] A 30-month programme shows daily-level tasks throughout, including month 30. The strongest inference is: - A. the planning team is unusually diligent - B. far-horizon detail is manufactured precision that will churn on every update, obscuring real variance - C. the schedule will need no re-baselining - D. rolling-wave planning has been correctly applied

Rationale: Detail beyond the knowable horizon creates update churn and false confidence — the opposite of diligence in effect (A) and the opposite of rolling wave (D); constant churn makes re-baselining more likely, not less (C).

MCQ 6.3-E [6.3.1 · Application] D needs 3 crews in weeks 9–15 and concurrent staging needs 2; the site cap is 4. The excess demand the histogram must clear is: - A. 1 crew for 7 weeks - B. 5 crews for 7 weeks - C. 2 crews for 7 weeks - D. nothing — $3 + 2 = 5$ is within a 4-crew cap across two activities

Rationale: Peak demand 5 against cap 4 leaves one excess crew-week in each of the seven overlap weeks — the precise quantity smoothing must relocate. B is total demand, not excess; C is the staging demand itself; D misreads the cap as per-activity when it is per-site.

MCQ 6.3-C [6.3.1 · Application] A hard 3-crew cap forces either a 4-week project extension (delay cost USD 45,000/week) or a second-shift waiver at USD 20,000/week for the same 4 weeks (end date held). The value-maximising choice and its saving are: - A. extend: saves USD 100,000 - B. second shift: saves USD 100,000 - C. second shift: saves USD 25,000 - D. they cost the same

Rationale: Extension costs $4 \times 45,000 = 180,000$; the shift premium $4 \times 20,000 = 80,000$ — choosing the shift saves USD 100,000 and keeps the date. A picks the dearer plan; C compares one week only; D ignores the 65,000-per-week difference... which compounds four times.

MCQ 6.3-D [6.3.3 · Recall] A legitimate planning package in a rolling-wave schedule must carry: - A. daily-level task detail for its whole span - B. a ranged duration consistent with its estimate class and a dated elaboration event under change control - C. zero float, to keep pressure on the team - D. a pinned finish date agreed with the sponsor

Rationale: Far-wave honesty is ranged duration plus a governed elaboration point. A is the manufactured precision rolling wave exists to avoid; C fabricates criticality; D is the typed milestone anti-pattern (Case study B).

■ SELF-CHECK — KA 6.3

1. Why is smoothing "spending the risk buffer"? — It consumes float, and float is the project's absorption capacity for the unknown (KA 6.2.2).
2. What makes a milestone honest? — Computed date, named owner, unambiguous done-definition, traceable logic.
3. What must a planning package carry to be legitimate? — A ranged duration consistent with its estimate class and a dated elaboration event under change control.

KNOWLEDGE AREA 6.4

Delivery flow: predictive, agile and hybrid; recovery and forecasting

Topics: 6.4.1 flow across delivery modes · 6.4.2 schedule compression and recovery · 6.4.3 forecasting and scenario analysis.

6.4.1 Flow across delivery modes

Predictive scheduling (this domain so far) plans the whole network and measures variance against baseline. **Agile** delivery (Domain 13 in full) fixes cadence and capacity and lets scope flow: the schedule questions become throughput (items per iteration), cycle time and forecast by velocity. **Hybrid** programmes — the commonest reality — run engineered streams predictively and product streams by cadence, joined at **integration milestones** that appear in the CPM network as fixed-capacity deliveries. The leader's job is translation: a velocity forecast ("the reporting module completes in 4–6 sprints") enters the network as a ranged duration on the integration activity, and the network returns what the programme needs from the agile stream — the latest acceptable delivery, read from **LS**. Neither mode is senior; the network prices time, the cadence protects focus, and governance (Domain 3, KA 3.1.3) holds both to evidence.

6.4.2 Schedule compression and recovery

The two levers. **Crashing** buys duration with money (more resources, overtime, expediting); **fast-tracking** buys it with risk (overlapping activities that logic preferred in sequence). Both obey the same law: work the critical path, and re-run the passes after every move, because **the critical path migrates**.

WORKED EXAMPLE

Worked example 6.4.2 — the crash that stopped paying.

1. **Setup.** Auriga's client offers a bonus of USD 45,000 ($\approx$ SAR 168,750) per week saved. Expediting C costs USD 30,000 per week, up to two weeks. Should the leader buy one week, two, or none?
2. **Formula.** Crash only while (weeks actually saved $\times$ value per week) $>$ crash cost — checked against the re-run passes, not the original network.
3. **Substitution.** Crash C by 1 (8 $\rightarrow$ 7): duration 25 $\rightarrow$ 24; net gain $45,000 - 30,000 = +15,000$. Crash C by 2 (8 $\rightarrow$ 6): path B-D-E-F (2+6+7+5+4) is now co-critical at 24 — duration stays 24; second week's net $0 - 30,000 = -30,000$.
4. **Result. Crash one week only:** +USD 15,000. The second week is pure loss.
5. **Interpretation.** Compression is subject to sharply diminishing returns because every crash promotes the next-longest path. The disciplined sequence: identify the binding path $\rightarrow$ price the cheapest week on it $\rightarrow$ re-run the passes $\rightarrow$ repeat until the next week costs more than it is worth. Leaders who mandate "crash procurement by two weeks" from the original network pay for weeks that no longer exist.

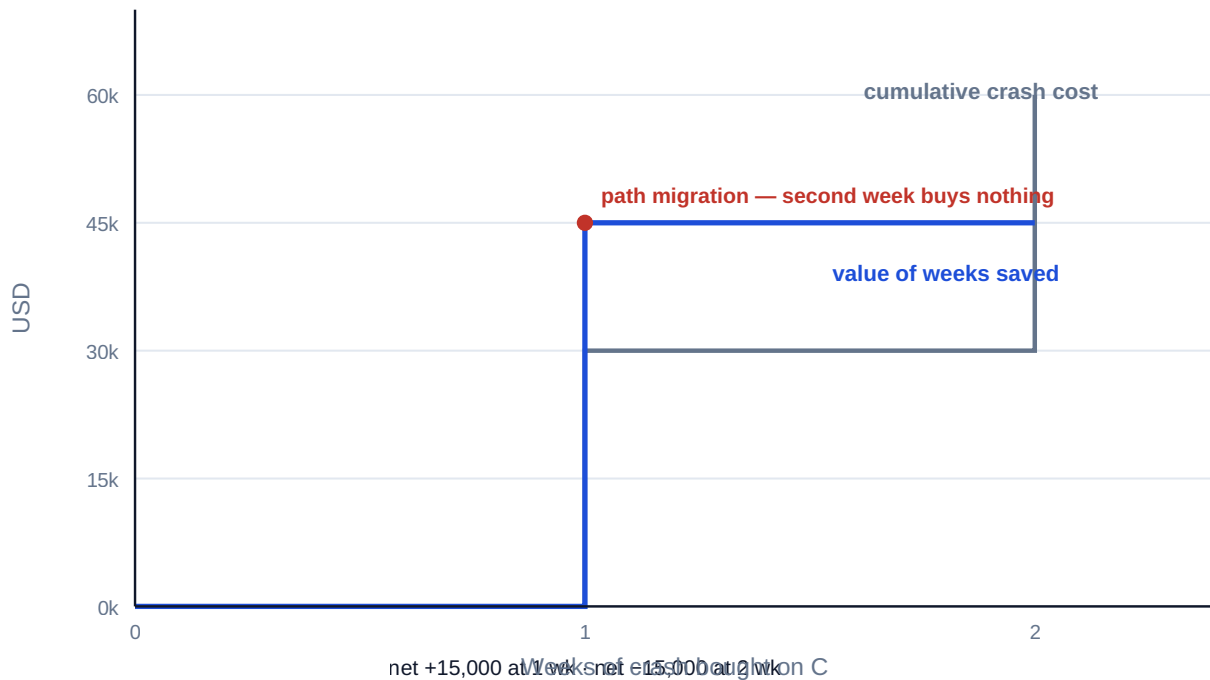


Fig 6.4.1 — The economics of crashing Auriga's procurement

Recovery. When the network slips (self-check 6.2.3: D at 10 weeks → 27), recovery options rank by cost-of-time: re-sequencing and logic re-choice (cheapest — KA 6.1's chosen logic); float harvesting from non-critical paths (G's 8 weeks fund nothing on the new critical path, but its resources might); crashing the new critical path (now through D); fast-tracking E against D's tail with an explicit rework risk (Domain 8's risk register prices it); and scope or acceptance re-negotiation (Domain 5) as the honest last resort. A **recovery plan** states the target date, the moves, their costs and risks, and the decision authority spending them — the template is toolkit 6.T.2.

6.4.3 Forecasting and scenario analysis

Three-point estimates. Uncertain durations are ranges, not points. The PERT expected duration and spread for an activity estimated optimistic o , most-likely m , pessimistic p :

$$t_e = (o + 4m + p) / 6 \quad \sigma = (p - o) / 6$$

WORKED EXAMPLE**Worked example 6.4.3 — Auriga's installation risk.**

1. **Setup.** Installation E is estimated $o = 4$, $m = 5$, $p = 12$ weeks (the long tail: a legacy-system integration problem discovered late).
2. **Formula.** $t_e = (o + 4m + p)/6$; $\sigma = (p - o)/6$.
3. **Substitution.** $t_e = (4 + 20 + 12)/6 = 36/6$; $\sigma = (12 - 4)/6 = 8/6$.
4. **Result.** $t_e = 6.0$ weeks (not 5); $\sigma = 1.33$ weeks.
5. **Interpretation.** The deterministic network used 5 weeks; the risk-weighted expectation is 6 — the skewed tail alone adds a week to the honest forecast, pushing expected completion toward 26 weeks before any risk event "happens". Deterministic dates are the mode, not the mean; boards deserve the mean and the spread. Full quantitative schedule risk analysis — correlating activities, simulating paths, criticality indices — is Domain 8 (KA 8.2), built on exactly this input.

Scenario analysis. Between the single date and the full simulation sits the leader's workhorse: three coherent scenarios (base / threat / opportunity), each a complete re-run of the passes under stated assumptions, each with owners for the assumptions that differ. Scenarios are cheap, auditable, and — because each is a real network — they expose path migration that a percentage-confidence number hides.

The earned-schedule bridge. Once cost joins schedule (Domain 7), progress data yields a time-based forecast: **earned schedule ES** asks when the value now earned was planned to have been earned, and $SPI(t) = ES / AT$ (actual time) measures schedule efficiency in time units. If Auriga's week-22 status shows work that the baseline expected by week 20, $SPI(t) = 20/22 = 0.91$ — the programme is running at 91 % of planned tempo, a signal the currency-based **SPI** famously loses late in a project. Domain 7 (KA 7.3) builds the full machinery; it is flagged here because schedule forecasting belongs to whoever owns the network, not only to the cost engineer.

AI in this KA

Machine forecasting earns trust in exactly one way: calibration. A model that ingests progress data and produces completion forecasts must show its record — forecast vs actual across past periods — before its numbers enter a board pack (Domain 14, KA 14.3's verification workflow). The leader's questions are the auditor's: what data trained it, what does it do when the data thins, who re-ran its arithmetic, and what would make it wrong? Fluent confidence without a calibration record is the machine version of the site manager who is "sure it'll be fine".

■ KEY TERMS — KA 6.4

Term	Meaning
Crashing / fast-tracking	Buying duration with money / with overlap risk.
Path migration	The critical path moving as durations change; why passes are re-run.
Recovery plan	Target date, ranked moves, costs, risks, decision authority.
Three-point / PERT estimate	$t_e = (o+4m+p)/6$, $\sigma = (p-o)/6$.
Scenario analysis	Complete network re-runs under coherent stated assumptions.
Integration milestone	The CPM node where an agile stream's delivery enters the network.

■ SAMPLE MCQS — KA 6.4

MCQ 6.4-A [6.4.2 · Application] Saving a week is worth USD 45,000; crashing the critical activity costs USD 30,000 per week for up to two weeks; after one week of crashing, a parallel path becomes co-critical. The value-maximising decision is: - A. crash two weeks (net +USD 30,000) - B. crash one week (net +USD 15,000) - C. crash nothing (avoid all cost) - D. crash two weeks and fast-track the parallel path at no cost

Rationale: Week one converts to a project week: +15,000. Week two is absorbed by the co-critical path: -30,000. A prices weeks that don't materialise; C leaves +15,000 unclaimed; D invents a free fast-track — overlap always carries rework risk.

MCQ 6.4-B [6.4.3 · Application] An activity is estimated $o = 4$, $m = 5$, $p = 12$ weeks. Its PERT expected duration is: - A. 5.0 weeks - B. 7.0 weeks - C. 6.0 weeks - D. 6.5 weeks

Rationale: $(4 + 4 \times 5 + 12)/6 = 36/6 = 6.0$. A is the most-likely value (the mode); B is the unweighted mean of the three points; D miscounts the weighting.

MCQ 6.4-C [6.4.1 · Analysis] A hybrid programme needs a software module from an agile team for an integration milestone. The schedule-sound way to join the two worlds is: - A. impose the CPM date on the team as a sprint deadline - B. enter the team's velocity-based forecast as a ranged duration and read the latest acceptable delivery from the backward pass - C. exclude the module from the network since agile work cannot be scheduled - D. convert the agile team to predictive planning for the integration period

Rationale: B translates in both directions — evidence-based forecast in, latest-start requirement out. A dictates without evidence; C leaves the network blind at a convergence point; D destroys the team's delivery system to decorate the network.

MCQ 6.4-F [6.4.3 · Application] An activity is estimated optimistic 3, most-likely 4, pessimistic 8 weeks. Its PERT expected duration and standard deviation are: - A. $t_e = 4.5$, $\sigma = 0.83$ - B. $t_e = 4.0$, $\sigma = 0.83$ - C. $t_e = 5.0$, $\sigma = 1.67$ - D. $t_e = 4.5$, $\sigma = 5.0$

Rationale: $t_e = (3 + 16 + 8)/6 = 4.5$; $\sigma = (8 - 3)/6 = 0.83$. B reports the mode as the mean; C is the unweighted three-point average with a doubled spread; D confuses σ with the pessimistic-minus-optimistic range.

MCQ 6.4-D [6.4.3 · Application] At week 22, a programme has earned the value its baseline planned to earn by week 20. Its time-based schedule performance index $SPI(t)$ is: - A. 1.10 - B. 0.91 - C. 0.80 - D. 20.0

Rationale: $SPI(t) = ES/AT = 20/22 = 0.91$ — the programme delivers at 91 % of planned tempo. A inverts the ratio; C subtracts the two weeks from the wrong base; D reports earned schedule itself, not the index.

MCQ 6.4-E [6.4.3 · Analysis] Compared with quoting "78 % confidence of the date", giving the board three fully re-run schedule scenarios is stronger because: - A. three numbers always beat one - B. each scenario is an auditable network with named assumptions, showing how the date moves and where the path migrates - C. percentages are always statistically invalid - D. scenarios eliminate the need for risk analysis

Rationale: The scenario's power is auditability and mechanism — assumptions with owners, visible path migration. A is numerology; C overclaims (a calibrated percentage is legitimate, Domain 8); D reverses the relationship — scenarios are inputs to risk analysis, not substitutes.

■ SELF-CHECK — KA 6.4

1. Why does compression show diminishing returns? — Each crash promotes the next-longest path; saved weeks stop converting to project weeks (path migration).
2. What makes a scenario more useful to a board than a confidence percentage? — It is a complete, auditable network with named assumptions — it shows how the date moves, not just that it might.
3. When may a machine forecast enter a board pack? — With a calibration record, verified arithmetic, stated data lineage and a named human owner.

— Advanced topics — Domain 6

6.A.1 The drum: constraint-aware flow

Where one resource is the system's constraint (a single commissioning team, one test rig), throughput scheduling subordinates everything to that drum: buffer it, never starve it, and let non-constraint efficiency go. This reframes utilisation worship — a 100 %-busy non-constraint builds inventory, not progress. In schedule terms: protect the constraint's feed path with deliberate float placement rather than spreading buffer evenly.

6.A.2 Schedule baselines and the update cycle

The control schedule lives on a cycle: status (actuals and remaining durations in), re-run passes, variance vs baseline, forecast, and — rarely, formally — re-baseline under Domain 4's change control. Two integrity rules carried from the family's controls tradition: never edit history (actualised dates are records), and never re-baseline to hide variance — a re-baseline is a governance decision with an audit trail, not a cosmetic one.

6.A.3 PDM edge cases: float under SS and FF links

Precedence-diagram (PDM) links change how float behaves, and two edge cases bite reviewers. **(1) The SS-linked short successor.** P (10 weeks) **SS+2** Q (3 weeks): Q can run weeks 2–5, finishing seven weeks before its predecessor — legal, and correct if the dependency truly binds only the starts; but if Q also needs P's output to finish, the network is missing an FF link, and Q's float is fiction. Paired **SS+FF** links express "starts together, finishes together" honestly. **(2) Duration-driven float.** Under an FF link the successor's float depends on its own duration: lengthen Q and its **LS** moves earlier — planners who "add duration for safety" on FF-linked activities silently destroy their float. The audit habit: on every SS/FF link, ask which end of each activity the logic genuinely binds, and whether the unbound end needs its own link. A network is only as honest as its least-considered link type.

6.A.4 Negative float and the honest schedule

A constraint-bound network can compute **TF < 0**: the plan, as constrained, is late already. Negative float is information — the size of the recovery problem — and suppressing it (deleting constraints, shortening durations by fiat) is the scheduling equivalent of cooking the books. The professional response mirrors KA 6.4.2: quantify it, trace the binding path, price the recovery options, and escalate the decision to whoever owns the trade (Domain 3's escalation design).

WORKED EXAMPLE**Worked example 6.A.4 — buying back two weeks of negative float.**

1. **Setup.** A newly imposed outage constraint requires Auriga to finish by **week 23**; logic says 25 — so $TF = -2$ on A-B-C-E-F. Available buys: crash C at USD 30,000/week (max 2); crash E at USD 55,000/week (max 1). Find the cheapest feasible recovery to week 23.
2. **Formula.** Recover week by week on the currently binding path(s), re-running the passes after each buy (path migration, 6.4.2).
3. **Substitution.** Week one: crash C by 1 → duration 24, but B-D-E-F is now co-critical at 24. Week two must shorten both paths: the only shared activity available is E → crash E by 1 (USD 55,000) → duration 23. (C's second week alone would leave B-D-E-F at 24 — spent money, no schedule.)
4. **Result.** Feasible recovery: **crash C × 1 + crash E × 1 = USD 85,000**, finishing week 23.
5. **Interpretation.** Negative float is bought back on the binding path as it migrates — the second-cheapest week on the original critical path (C's second week, USD 30,000) is worthless, while the dearest activity (E, on both paths) is the only week-two purchase that works. This is why recovery plans list options in re-run order, not in unit-cost order (toolkit 6.T.2).

— Industry variations — Domain 6

The passes are universal; what counts as a schedule, and what discipline it needs, is sectoral:

- **Construction and EPC.** Deep L3/L4 hierarchies, thousand-activity networks, contractual float-ownership clauses (Domain 10) and delay-analysis protocols make float a legal quantity — the honesty settings of the Executive perspective are contract compliance here, not just culture.
- **Technology and product.** Hybrid is the default (KA 6.4.1): cadence-based streams joined to a thin predictive network at integration milestones; the commonest defect is a network so sparse that convergence points are invisible until they slip.
- **Shutdowns and turnarounds.** The network runs in hours; near-critical analysis widens to every path within minutes of critical, and the drum logic of 6.A.1 governs a single shared crane or permit desk. Compression economics (6.4.2) turn over in shifts, not weeks.
- **Pharmaceutical and regulated development.** Milestones are regulatory events with submission-window physics — genuine date constraints (6.3.2), modelled as logic, with the pin-and-squeeze temptation at its most dangerous because windows are unforgiving.
- **Public programmes.** Announced dates precede networks (Case study B); the leader's protection — milestones only after the backward pass — is hardest exactly where it matters most, and negative-float honesty (6.A.4) is the difference between an early re-plan and a public failure.

— Case study — Domain 6: recovering Auriga (utilities / technology)

Situation. End of week 13. Civil and cabling works D — planned at 7 weeks from week 8 — hits contaminated-ground remediation; the discipline lead's honest re-estimate takes D to 10 weeks (finish week 18). The client's outage window for final commissioning opens at week 25 and closing the window costs USD 45,000 per week of delay. The team re-runs the passes.

Analysis. With $D = 10$: E starts week 18 (D now later than C's week 16), F finishes week 27 — **two weeks into the penalty window**, and the critical path has migrated to **A-B-D-E-F**. Procurement C, critical all project, now carries $TF = 2$. G still holds float.

The recovery decision. Options priced by the team: (1) crash the old critical path — worthless, C is no longer binding; (2) crash D's remediation with a second civil crew — USD 35,000 for one week; (3) fast-track E against D's tail — start installation in the uncontaminated sections at week 16, accepted rework risk assessed at $20\% \times \text{USD } 60,000 = \text{USD } 12,000$ expected cost, one further week saved; (4) do nothing — $2 \times 45,000 = \text{USD } 90,000$ penalty exposure. The leader takes (2) + (3): cost $\approx \text{USD } 47,000$ expected against USD 90,000 — and finishes at week 25 with the window intact.

What the domain teaches here. Re-run the passes before spending a cent (the instinctive "crash procurement" would have bought nothing); price recovery in expected cost including risk; harvest overlap where the physics genuinely allows it; and report the new float positions to the client honestly — the schedule survived because the remediation estimate was honest a full five weeks before the window (Domain 11's reporting culture, applied to time).

— Case study B — Domain 6: the milestone that was typed, not computed (public programme)

Situation. A government services programme publishes a go-live milestone eighteen months out, set in a ministerial announcement before the L3 schedule existed. When the network is finally built, the backward pass puts the milestone's required start for user-acceptance testing three weeks before the environment that testing needs can exist — $TF = -3$ from day one.

What happened. For six reporting cycles the programme's L1 view showed the milestone green: the date was pinned, and each slip downstream was absorbed by silently shortening the testing window — the classic pin-and-squeeze. In cycle seven an assurance review (Domain 3, KA 3.3) ran the passes without the pin, surfaced the negative float, and forced the choice the pin had deferred: descope the first release (Domain 5), or move the date. The minister moved the date — at a political cost several times what a computed date would have cost eighteen months earlier.

What the domain teaches here. A typed milestone is a promise without a plan. Negative float is not embarrassment to be formatted away; it is the earliest, cheapest warning the programme will ever get (6.A.4). And the leader's protection is procedural, not heroic: milestones enter public commitments only after the backward pass says they exist.

— Executive perspective — Domain 6

What a project leader cannot delegate in this domain:

- **The logic sign-off.** Discipline leads own their links; the leader owns the assembled claim that this network is how the project actually works.
- **Float policy.** Who may spend float, in what increments, reported how — decided before the first progress update, not during the first crisis.
- **The honesty settings.** No pins doing logic's work; negative float reported the cycle it appears; re-baselining only through governance. Schedules fail morally before they fail mathematically.

- **The compression chequebook.** Crash and fast-track decisions are investments under uncertainty — the leader signs the expected-value arithmetic (6.4.2) and the risk acceptance (Domain 8), never a bare instruction to "go faster".
- **The translation duty.** Boards hear dates; teams live in passes, floats and scenarios. The leader is the honest converter between the two languages — both directions.

— Calculation exercises — Domain 6

Exercise 6.1 Using the Auriga network, compute **ES/EF** for every activity and the project duration, showing the convergence at E. Solution. A 0-2 · B 2-8 · C 8-16 · D 8-15 · E $\max(16,15)=16$ -21 · F 21-25 · G 15-17. Duration **25 weeks**; E's start is set by C (16), not D (15) — the convergence rule takes the later predecessor. Common error: taking the earlier predecessor at a merge (giving E 15-20 and a false 24-week project).

Exercise 6.2 Complete the backward pass and state **TF** and **FF** for D and G. Solution. From **LF** = 25: F 21/25 · E 16/21 · C 8/16 · G 23/25 · D **LF** = $\min(E \text{ LS } 16, G \text{ LS } 23) = 16$, **LS** = 9. **D: TF = 1, FF = $\min(16,15) - 15 = 0$. G: TF = 8, FF = 25 - 17 = 8**. Common error: computing D's **FF** against only E (giving 1) — free float tests every successor.

Exercise 6.3 The client offers USD 45,000 per week saved; crashing C costs USD 30,000/week (max 2 weeks); crashing E costs USD 55,000/week (max 1 week). Find the value-maximising plan. Solution. Crash C by 1: 25→24, +15,000. Second week of C: co-critical B-D-E-F holds 24 — -30,000, reject. Crash E by 1 (E is on both paths): 24→23, **45,000 - 55,000 = -10,000**, reject. **Plan: crash C by one week; net +USD 15,000**. Common error: crashing E first because it is "on more paths" — path count doesn't change its negative unit economics.

Exercise 6.4 Activity durations: o = 6, m = 8, p = 16 weeks. Compute **t_e** and **σ**, and state the deterministic bias. Solution. **t_e = (6 + 32 + 16)/6 = 9.0 weeks**; **σ = (16 - 6)/6 = 1.67 weeks**. The deterministic plan at m = 8 understates the expectation by a full week — right-skewed tails always pull the mean above the mode.

Exercise 6.5 A programme's network shows **TF** = -2 on the binding path. List, in cost order, the recovery families and the governance step each requires. Solution. (1) Logic re-choice/re-sequencing — planner's proposal, leader's sign-off; (2) float and resource harvesting from non-binding paths — float-policy authority; (3) crashing the binding path — expected-value case, budget authority; (4) fast-tracking — risk acceptance recorded in the register; (5) scope/acceptance renegotiation — sponsor and change control. Reporting the -2 honestly precedes all five (6.A.4).

— Practitioner's toolkit — Domain 6

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 6.T.1 — Schedule quality gate (run before trusting any network)

- [] Zero dangles; zero loops; every lag has a stated reason.

- [] Date constraints listed, each justified as genuine external physics — none doing logic's job.
- [] Durations owned by the executing discipline, estimate class stated (far waves: ranged).
- [] Passes re-run after every change; critical-path durations sum exactly to project duration.
- [] Near-critical paths (within 10 %) listed with their floats.
- [] Float report distinguishes **TF** and **FF**; float policy names who may spend it.
- [] Milestones computed, owned, done-defined; no typed dates.
- [] AI-drafted logic or forecasts marked, verified, and owned by a named human.

Toolkit 6.T.2 — Recovery plan (one page)

Slip statement (activity, cause, size, date detected) · re-run pass results (new duration, new critical path, float table) · options priced in expected cost (re-sequence / harvest / crash / fast-track / renegotiate), each with risk and owner · selected plan and its decision authority · revised commitments and the stakeholder notice list · review date.

Toolkit 6.T.3 — Milestone register

Per milestone: computed date and current **TF** · owner · done-definition (acceptance reference) · contractual status and counterparty · L3 logic trace · reporting treatment (internal / contractual / public) · change history.

— Exam preparation — Domain 6

The calculation traps. Taking the earlier predecessor at a convergence (Exercise 6.1) · computing **FF** against one successor instead of all (Exercise 6.2) · quoting **TF** after it has been spent · crashing from the original network instead of re-running passes (path migration, 6.4.2) · treating the PERT mode as the mean (6.4.3) · reading a pinned milestone as a forecast · confusing project-level float with activity-level slack in reports.

Reflection questions. 1. Your programme's board pack shows every milestone green. What three questions establish whether that is information or formatting? (Computed or typed? Passes re-run this cycle? Where is the float table?) 2. A subcontractor asks for "the float" on their package. What must you settle before answering? (Which float — **TF** or **FF**; the float policy; contractual float ownership — Domain 10.) 3. When did you last see money spent on compression that the passes would have shown was worthless — and which governance step was missing? (6.4.2; toolkit 6.T.2.)

— Domain 6 summary

This domain turned "when will it finish?" from an opinion into an instrument. Planning levels give each audience a true view of one logic; dependency discipline makes the network a model rather than a drawing; the forward and backward passes yield the dates, the two floats and the critical path — with the master project showing why free float can vanish while total float survives, and why convergence points compound slippage. Resources, constraints, milestones and rolling waves connect the network to the real world's crews, windows and horizons. Delivery flow closes the loop: compression priced by expected value and re-run passes, recovery ranked from re-choice to renegotiation, PERT and scenarios replacing single dates with honest ranges, and agile streams joined to the network through ranged integration milestones. The leadership spine throughout: schedules fail morally before they fail mathematically — computed dates, visible float, reported

negative float, and machine output verified by named humans are what make a programme's word worth something.

07

DOMAIN 7

Cost, Resources and Commercial Awareness (quantitative flagship)

Group: Delivering the work (Domain 7 of 6 in Part Two — the cost counterpart to Domain 6's schedule).

Target: ~78 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries ([docs/books/registries/](#)). This domain is this book's home of the earned-value symbols — **PV**, **EV**, **AC**, **BAC**, **CV**, **SV**, **CPI**, **SPI**, **EAC**, **ETC**, **VAC**, **TCPI** — carried unchanged from the PCI family master table. Note the family notation-clash rule: **PV** here is **Planned Value**; present value is written **PV(x)** or in words (PFL-AI, Domain 3). British English; USD (+SAR where useful, indicative **USD 1 ≈ SAR 3.75**).

— Why this domain exists

A leader who can defend a date but not a number is half-equipped. Domain 6 built the schedule; this domain builds the money that pays for it and the commercial arrangements that bind other people to deliver it. It starts with estimating and budgeting — how a credible number is constructed and what its accuracy claim actually means (KA 7.1); establishes the cost baseline and the forecasting that keeps it honest (KA 7.2); builds **earned value** in full, because it is the only technique that answers cost and schedule performance in one integrated language (KA 7.3); and closes with the commercial awareness a leader cannot delegate — resource economics, procurement strategy, contract models and cash flow (KA 7.4). Risk quantification deepens in Domain 8; contracts and supply networks get their own treatment in Domain 10. What belongs here is the leader's own numeracy: enough to know when a forecast is arithmetic and when it is hope.

Learning objectives. After this domain a candidate can: select an estimating method appropriate to the available definition and state its accuracy class; build a three-point cost estimate; assemble a cost baseline separating contingency from management reserve; compute and interpret the full earned-value set (**CV**, **SV**, **CPI**, **SPI**); forecast with each **EAC** method and choose the one matching the variance's cause; compute **VAC** and **TCPI** and explain what **TCPI** demands of the remaining work; read resource economics through blended rates; distinguish the main contract models by who carries cost risk; compute a cost-incentive fee and the point of total assumption; explain why cash flow and profit differ on a project; and govern AI-produced cost forecasts under the family verification rule.

The master worked project. Project Auriga continues from Domain 6 — the 25-week control-systems upgrade for a regional utility — now with money attached. Its approved cost baseline is **BAC = USD 4,000,000**. At the **data date, end of week 13**, the baseline says **PV = USD 2,080,000**; measurement gives **EV = USD 1,920,000** and **AC = USD 2,120,000**. Every calculation in KA 7.2–7.3 uses these four numbers.

KNOWLEDGE AREA 7.1

Estimating and budgeting

Topics: 7.1.1 estimating methods and accuracy classes · 7.1.2 three-point estimating · 7.1.3 from estimate to budget.

7.1.1 Estimating methods and accuracy classes

The principle. An estimate's accuracy is governed by how well the work is defined, not by how much effort went into the arithmetic. The three standard methods trade definition against speed:

Method	How it works	Needs	Typical use
Analogous (top-down)	Scale a comparable past project	A true analogue and a scaling basis	Concept, screening (Domain 2)
Parametric	Cost = rate × parameter (per metre, per point, per kW)	A calibrated rate from real history	Early design, repetitive work
Bottom-up	Estimate each work package, then sum	A WBS at package level (Domain 4)	Baseline setting, control

Accuracy classes. Mature cost practice publishes an estimate's class alongside its number — a range that narrows as definition matures (the AACE Total Cost Management framework's class-5-to-class-1 progression is the reference treatment, described here in this book's own words). The professional discipline is simple and widely broken: **an estimate is never a single number without a range and a class.** "USD 4 million" is not an estimate; "USD 4 million, −15 %/+30 %, at a class consistent with 30 % design completion" is.

Common pitfall — precision mistaken for accuracy. A bottom-up estimate summing 400 packages to USD 4,183,662 looks authoritative and is no more accurate than its worst assumption. Leaders should be more suspicious of an over-precise number than a rounded one.

7.1.2 Three-point estimating

Cost, like duration (Domain 6, KA 6.4.3), is a distribution. The same PERT weighting applies:

$$C_e = (o + 4m + p) / 6 \quad \sigma = (p - o) / 6$$

WORKED EXAMPLE**Worked example 7.1.2 — Auriga's control-hardware package.**

1. **Setup.** Procurement of control hardware is estimated **optimistic USD 680,000, most-likely USD 750,000, pessimistic USD 1,000,000** (the tail: a single-source controller with volatile lead-time pricing).
2. **Formula.** $C_e = (o + 4m + p)/6$; $\sigma = (p - o)/6$.
3. **Substitution.** $C_e = (680,000 + 3,000,000 + 1,000,000)/6 = 4,680,000/6$; $\sigma = 320,000/6$.
4. **Result.** $C_e = \text{USD } 780,000$; $\sigma \approx \text{USD } 53,333$.
5. **Interpretation.** The most-likely figure is 750,000, but the expected cost is 780,000 — the right-skewed tail adds USD 30,000 before anything goes wrong. Budgeting at the mode systematically under-funds a portfolio of such packages; the difference between mode and mean, summed across a project, is a large part of what contingency exists to cover (7.1.3, and Domain 8's quantification).

7.1.3 From estimate to budget

The structure. An estimate becomes a controllable budget through a deliberate hierarchy:

work-package estimates	
→ control-account budgets	(WBS × organisation – the management-control points)
→ + contingency reserve	→ the COST BASELINE (BAC): the PM's authority
→ + management reserve	→ the total project funding requirement

Two rules carry the discipline. **Contingency is inside the baseline**, sized against identified risks (Domain 8) and spent by the project manager under a stated protocol. **Management reserve is outside the baseline**, held for unknown-unknowns, and released only by the sponsor or change authority (Domain 3's decision rights) — releasing it changes the baseline through change control (Domain 4, KA 4.4). Blurring the two is how projects appear to be "on budget" while consuming their own funding runway.

Time-phasing. The baseline is spread across the schedule to produce the **cumulative cost curve** (the S-curve) — and that curve is Planned Value (KA 7.3). A budget with no phasing cannot be measured against; the schedule (Domain 6) is therefore a precondition of cost control, not a parallel activity.

AI in this KA

Estimating is where AI assistance is most seductive and most in need of provenance. A model can produce a plausible parametric rate instantly, and it cannot tell you whether that rate came from comparable work, a different market, or nowhere at all. The governed workflow: **AI proposes** a method, a rate and a range; the estimator supplies the calibration evidence (which projects, which years, which escalation basis — the source discipline of the shared registry); the range and class are stated; and a named human owns the number. **AI proposes; the professional verifies, decides and remains accountable.** An estimate whose basis cannot be produced on request is not an estimate.

■ KEY TERMS — KA 7.1

Term	Meaning
Analogous / parametric / bottom-up	Scale an analogue · rate × parameter · sum the packages.
Accuracy class	The definition-linked range accompanying an estimate; never optional.
Control account	The WBS × organisation point where scope, budget and actuals integrate.
Contingency reserve	Inside the baseline; funds identified risks; PM-controlled.
Management reserve	Outside the baseline; unknown-unknowns; sponsor-controlled.
BAC	Budget at Completion — the time-phased cost baseline's total.

■ SAMPLE MCQS — KA 7.1

MCQ 7.1-A [7.1.2 · Application] A package is estimated o = 680,000, m = 750,000, p = 1,000,000. Its PERT expected cost is: - A. USD 750,000 - B. USD 780,000 - C. USD 810,000 - D. USD 840,000

Rationale: $(680,000 + 4 \times 750,000 + 1,000,000)/6 = 780,000$. A is the mode; C is the unweighted three-point mean; D over-weights the pessimistic value.

MCQ 7.1-B [7.1.3 · Analysis] A project reports "on budget" while having consumed 60 % of its management reserve at 40 % complete. The correct reading is: - A. genuinely on budget — management reserve exists to be spent - B. the baseline is intact but the project's total funding is eroding faster than progress; the trend belongs in the next report to the sponsor - C. a baseline breach requiring immediate re-baselining - D. an accounting error, since management reserve is inside the baseline

Rationale: Management reserve sits outside the baseline (so D is wrong and A is technically true but misleading), and its release is a sponsor-level signal, not a project-level convenience. It is not yet a breach (C) — it is the early warning that precedes one.

MCQ 7.1-C [7.1.1 · Recall] Which statement about a bottom-up estimate summing to USD 4,183,662 is soundest? - A. its precision indicates high accuracy - B. its accuracy is bounded by its assumptions and definition maturity, and it must still carry a range and class - C. rounding it would reduce its accuracy - D. bottom-up estimates do not need accuracy classes

Rationale: Precision (digits) and accuracy (closeness to outturn) are independent; definition maturity governs the latter. Rounding changes no information (C), and every estimate carries a class (D).

■ SELF-CHECK — KA 7.1

1. Where do contingency and management reserve sit, and who spends each? — Contingency inside the baseline, PM-controlled; management reserve outside it, sponsor-controlled.
2. Why is the time-phased baseline a precondition for cost control? — Because the phased cumulative curve is Planned Value; without it there is nothing to measure performance against.
3. What two things must accompany every estimate? — A range and an accuracy class tied to definition maturity.

KNOWLEDGE AREA 7.2

The cost baseline, actuals and forecasting

Topics: 7.2.1 measuring actual cost · 7.2.2 the forecasting question · 7.2.3 baseline integrity.

7.2.1 Measuring actual cost

The principle. AC must cover the same work as EV, in the same period, or every index built from them is fiction. Three mechanics decide whether it does:

- **Accruals.** Work received but not yet invoiced belongs in this period's AC. Omit accruals and cost performance looks excellent until the invoices land — the commonest cause of a "sudden" overrun that was months old.
- **Commitment vs actual.** A purchase order is a commitment, not a cost. Both matter: AC drives performance, commitments drive the funding forecast. Confusing them double-counts or hides money.
- **Open-commitment hygiene.** Stale purchase orders left open inflate the forecast; cleansing them is a standing month-end task.

7.2.2 The forecasting question

A forecast answers one question — what will this cost when it is done? — and the honest answer depends entirely on **why** the current variance exists. That judgment is the leader's; the arithmetic is KA 7.3.3's EAC family. The rule to internalise now: **a forecast is a statement about the remaining work, not an extrapolation ritual.** A variance caused by a closed, non-recurring event says nothing about what is left; a variance caused by a systemic productivity shortfall says everything.

7.2.3 Baseline integrity

The baseline's authority rests on its stability. Four standing rules: budgets of completed or open work packages are never retrospectively adjusted (except to correct an authorised error); re-baselining happens only through change control, with an audit trail (Domain 4, KA 4.3); transfers between control accounts are logged, not silent; and the baseline never absorbs a variance by moving budget to where the money went. A baseline edited to match reality has stopped measuring anything — the cost analogue of Domain 6's pinned milestone.

■ KEY TERMS — KA 7.2

Term	Meaning
AC	Actual Cost of the work performed, including period accruals.
Accrual	Cost of work received but not yet invoiced, recognised in the period.
Commitment	A contractual obligation (e.g. a PO); a funding fact, not yet a cost.
Re-baselining	A governed change to the baseline; never a variance-hiding device.

■ SAMPLE MCQS — KA 7.2

MCQ 7.2-A [7.2.1 · Analysis] A project's CPI has read 1.02 for four months; then one month it drops to 0.91 with no change in productivity. The likeliest explanation is: - A. genuine sudden

inefficiency - B. accruals were not being recognised, so earlier AC understated cost and this month absorbed the catch-up - C. the baseline was too generous - D. EV was over-claimed this month

Rationale: A step change in CPI without an operational change points at measurement, and missing accruals are the classic cause — earlier periods flattered, one period punished. D would raise, not lower, the earlier readings' credibility, and C would show as a stable, not stepped, pattern.

MCQ 7.2-B [7.2.3 · Recall] Which action preserves baseline integrity? - A. moving budget from an underspent control account to cover an overspend, without record - B. re-baselining through change control with an audit trail when scope genuinely changes - C. adjusting a completed package's budget to match its actual cost - D. reducing remaining budgets so the total still equals BAC

Rationale: Only governed change preserves the measurement. A is a silent transfer, C is retrospective adjustment, D is the classic "make the numbers add up" manoeuvre — each destroys comparability.

■ SELF-CHECK — KA 7.2

1. Why must AC include accruals? — So it covers the same work as EV in the same period; otherwise every index is misstated.
2. Is a purchase order a cost? — No: a commitment. It informs the funding forecast, not performance.
3. What single question does a forecast answer? — What the remaining work will cost, given why the variance exists.

KNOWLEDGE AREA 7.3

Earned value: measurement, variances and forecasting

Topics: 7.3.1 the three measures · 7.3.2 variances and indices · 7.3.3 the EAC family · 7.3.4 VAC and TCPI.

7.3.1 The three measures

Definitions. All three are expressed in the same budget currency so they are directly comparable:

- **Planned Value (PV)** — the budgeted cost of the work scheduled by the data date: the time-phased baseline of KA 7.1.3.
- **Earned Value (EV)** — the budgeted cost of the work actually performed: physical progress **valued at the budget rate**, never at what it cost.
- **Actual Cost (AC)** — the cost actually incurred for that work, accruals included (KA 7.2.1).

The single most important conceptual point: EV is measured at budget. That is what lets EV be compared with PV (both at budget → schedule progress) and with AC (budget vs actual for the same work → cost efficiency). Confusing "value earned" with "cost incurred" collapses the method.

Earning rules. How EV is claimed decides how much it can be gamed: **0/100** (nothing until complete — objective, good for short packages); **50/50; percent complete** (needs an objective basis); **units completed** (best where output is countable); **weighted milestones**; and **level of effort**, where EV is set equal to PV by the calendar, so it can never show a schedule variance. Level of effort dilutes whatever it is mixed into — formal practice segregates it and caps its share of a control account.

7.3.2 Variances and indices

$$\begin{array}{ll} CV = EV - AC & SV = EV - PV \\ CPI = EV / AC & SPI = EV / PV \end{array}$$

WORKED EXAMPLE

Worked example 7.3.2 — Auriga at week 13.

- Setup.** $BAC = 4,000,000$; at the data date $PV = 2,080,000$, $EV = 1,920,000$, $AC = 2,120,000$.
- Formula.** As above; variances in currency, indices as ratios (two decimals).
- Substitution.** $CV = 1,920,000 - 2,120,000$; $SV = 1,920,000 - 2,080,000$; $CPI = 1,920,000 / 2,120,000$; $SPI = 1,920,000 / 2,080,000$.
- Result.** $CV = (\text{USD } 200,000)$ · $SV = (\text{USD } 160,000)$ · $CPI = 0.91$ · $SPI = 0.92$. Auriga is **48.0 % complete** (EV/BAC) having spent **53.0 % of budget** (AC/BAC).
- Interpretation.** Behind and over: for every dollar spent the project is producing 91 cents of budgeted work, and it has delivered 92 % of what it planned to by now. The percent-complete versus percent-spent pair (48 vs 53) is the same story in the form a board absorbs fastest. Note this is the cost view of the very slippage Domain 6's case study recovered on the schedule side — the two domains are describing one project.



Fig 7.3.1 — Auriga's earned-value S-curves at week 13

7.3.3 The EAC family — forecasting

ETC = the remaining work's forecast cost	EAC = AC + ETC
(a) EAC = AC + (BAC - EV)	remaining work at the BUDGETED rate
(b) EAC = BAC / CPI	current cost efficiency PERSISTS
(c) EAC = AC + (BAC - EV) / (CPI × SPI)	cost AND schedule pressure compound
(d) EAC = AC + bottom-up ETC	re-estimate the remainder from scratch

WORKED EXAMPLE

Worked example 7.3.3 — the same project, four futures.

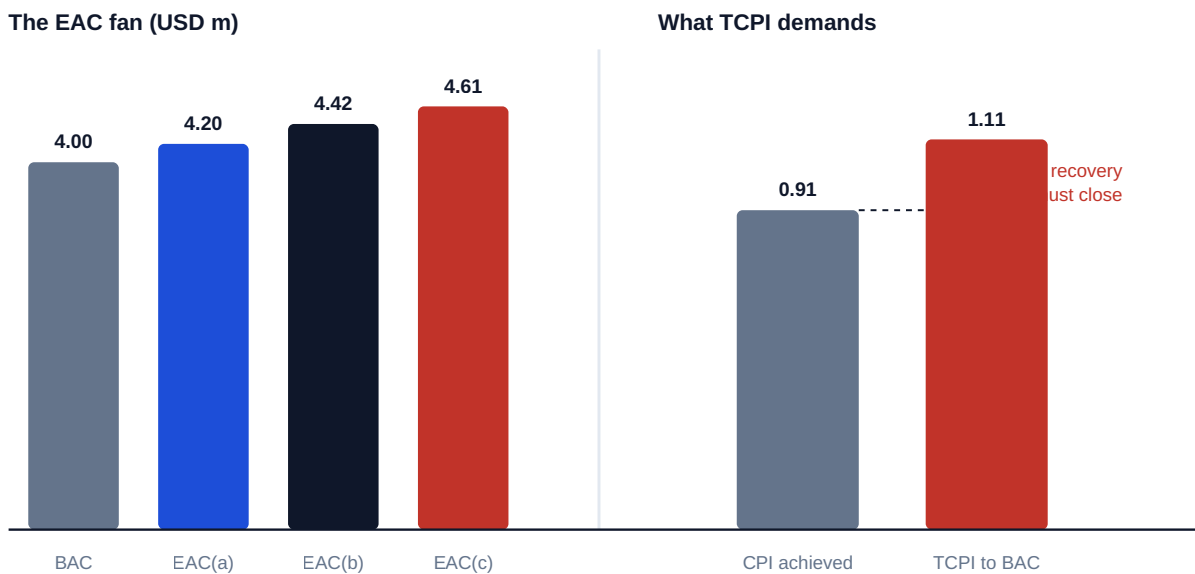
- Setup.** Auriga's week-13 figures, forecast to completion by each method.
- Formula.** (a)–(c) above; **CPI** = 0.905660, **SPI** = 0.923077 at full precision.
- Substitution.** (a) $2,120,000 + (4,000,000 - 1,920,000)$; (b) $4,000,000 / 0.905660$; (c) $2,120,000 + 2,080,000 / (0.905660 \times 0.923077)$.
- Result.** (a) **USD 4,200,000** · (b) **USD 4,416,667** · (c) **USD 4,608,056**.
- Interpretation.** The same four measured numbers support forecasts spanning **USD 408,056** — over 10 % of **BAC**. The spread is not uncertainty in the arithmetic; it is the assumption about the remaining work, made explicit. Choosing among them is the leader's judgment, and it must be stated: if Auriga's overrun was the contaminated-ground remediation of Domain 6 — a discrete, now-closed event — method (a) is right and the others over-forecast. If it reflects a systemic productivity shortfall that will persist, (b). If schedule pressure is now driving overtime and expediting, (c). A forecast presented without its assumption named is not a forecast; it is a number.

7.3.4 VAC and TCPI

VAC = BAC - EAC	the forecast overrun/underrun
TCPI = (BAC - EV) / (BAC - AC)	efficiency the REMAINING work must achieve to hit BAC
TCPI = (BAC - EV) / (EAC - AC)	...to hit a revised EAC

WORKED EXAMPLE**Worked example 7.3.4 — what recovery would actually require.**

1. **Setup.** Auriga's week-13 figures; test recovery to **BAC**, and to the method-(b) forecast.
2. **Formula.** $VAC = BAC - EAC$; **TCPI** as above.
3. **Substitution.** $VAC(a) = 4,000,000 - 4,200,000$; $VAC(b) = 4,000,000 - 4,416,667$. $TCPI(BAC) = 2,080,000 / (4,000,000 - 2,120,000) = 2,080,000/1,880,000$. $TCPI(EAC b) = 2,080,000 / (4,416,667 - 2,120,000)$.
4. **Result.** $VAC(a) = (\text{USD } 200,000)$ · $VAC(b) = (\text{USD } 416,667)$. **TCPI to BAC = 1.11**; **TCPI to the (b) forecast = 0.91**.
5. **Interpretation.** To finish at budget, the remaining 52 % of work must run at **1.11** — an 11 % efficiency gain by a team currently achieving 0.91. That is a 22-point swing, and the honest question in the room is what specifically would deliver it. Meanwhile **TCPI to the (b) forecast is 0.91** — exactly today's **CPI**, which is the arithmetic identity worth internalising: **forecasting with BAC/CPI is precisely the assumption that nothing changes**. **TCPI** is the reality check on every recovery promise: a required index far above demonstrated performance is a plan that needs a mechanism, not encouragement.

**Fig 7.3.2 — The 'EAC' fan and what 'TCPI' demands****AI in this KA**

Earned value is arithmetic on four numbers, so machine computation is safe and instant verification is cheap — which makes unverified AI output indefensible here rather than merely risky. The deterministic checks: EV/BAC must equal the claimed percent complete; $CV = EV - AC$ and $CPI = EV/AC$ recomputed independently; **TCPI to BAC/CPI** must equal **CPI** (7.3.4's identity); and every **EAC** must state its assumption. Where AI forecasts from trend data, the calibration rule of Domain 6 (KA 6.4) applies unchanged: show the record of past forecasts against outturn before the number enters a board pack, and name the human who owns it.

■ KEY TERMS — KA 7.3

Term	Meaning
PV EV AC	Budgeted cost of work scheduled · performed · the cost actually incurred.
CV SV	$EV - AC \cdot EV - PV$, in currency.
CPI SPI	$EV/AC \cdot EV/PV$, as ratios.
EAC ETC	Estimate at / to complete; $EAC = AC + ETC$.
VAC	$BAC - EAC$ — forecast variance at completion.
TCPI	Efficiency the remaining work must achieve to hit a stated target.
Level of effort	Earning by calendar ($EV \equiv PV$); can never show schedule variance.

■ SAMPLE MCQS — KA 7.3

MCQ 7.3-A [7.3.2 · Application] BAC 4,000,000; PV 2,080,000; EV 1,920,000; AC 2,120,000. CPI and SPI are: - A. 0.91 and 0.92 - B. 1.10 and 1.08 - C. 0.92 and 0.91 - D. 0.91 and 1.02

Rationale: $CPI = 1,920,000/2,120,000 = 0.91$; $SPI = 1,920,000/2,080,000 = 0.92$. B inverts both ratios; C swaps them (dividing EV by the wrong denominator); D miscomputes SPI against BAC-derived progress.

MCQ 7.3-B [7.3.3 · Analysis] The overrun was caused by a one-off ground-remediation event, now closed, and the remaining work is expected to run to budget. The appropriate EAC is: - A. $AC + (BAC - EV) = 4,200,000$ - B. $BAC/CPI = 4,416,667$ - C. $AC + (BAC - EV)/(CPI \times SPI) = 4,608,056$ - D. $BAC = 4,000,000$

Rationale: A discrete, closed cause makes the variance **atypical**, so remaining work is forecast at the budgeted rate. B assumes the inefficiency persists and C that it compounds with schedule pressure — both contradict the stated cause. D ignores money already spent above budget.

MCQ 7.3-C [7.3.4 · Application] With BAC 4,000,000, EV 1,920,000 and AC 2,120,000, the TCPI required to complete at BAC is: - A. 0.91 - B. 1.00 - C. 1.11 - D. 1.21

Rationale: $(4,000,000 - 1,920,000)/(4,000,000 - 2,120,000) = 2,080,000/1,880,000 = 1.11$. A is the demonstrated CPI; B assumes recovery needs only par performance; D uses PV in place of AC in the denominator.

MCQ 7.3-D [7.3.1 · Analysis] A control account is 70 % level-of-effort by budget. Its reported SPI of 1.00 most likely means: - A. the account is exactly on schedule - B. little about schedule: level of effort earns by the calendar, so $EV \equiv PV$ for most of the account regardless of progress - C. the discrete work is ahead, offsetting a delay - D. the earning rules were misapplied

Rationale: LOE sets EV equal to PV by construction, so a heavily-LOE account reads 1.00 whatever happens — which is why practice segregates and caps it. C invents an offset the data cannot show; the rules may have been applied entirely correctly (D), and that is the problem.

■ SELF-CHECK — KA 7.3

1. Why is EV measured at budget, not at cost? — So it is comparable with PV (schedule) and with AC (efficiency); at cost it would collapse into AC.
2. What does TCPI = 1.11 against CPI = 0.91 tell a leader? — Recovery to budget requires a 22-point efficiency swing; the plan needs a named mechanism, not optimism.

3. State the identity linking *TCPI* and *BAC/CPI*. — *TCPI* to an *EAC* of *BAC/CPI* equals the current *CPI*: that forecast is the assumption that nothing changes.

KNOWLEDGE AREA 7.4

Resource economics, procurement strategy and cash

Topics: 7.4.1 resource economics and blended rates · 7.4.2 contract models and cost risk · 7.4.3 incentive fees and the point of total assumption · 7.4.4 cash flow versus profit.

7.4.1 Resource economics and blended rates

Most project cost is people. A leader who reasons in headcount rather than **cost per unit of capability** will mis-price every option in Domain 6's compression decisions.

WORKED EXAMPLE

Worked example 7.4.1 — Auriga's engineering blended rate.

- Setup.** The engineering pool is 40 technicians at USD 95/hour, 25 engineers at USD 140/hour, 15 senior specialists at USD 210/hour.
- Formula.** Blended rate = $\Sigma(\text{count} \times \text{rate}) / \Sigma(\text{count})$.
- Substitution.** $(40 \times 95 + 25 \times 140 + 15 \times 210) / 80 = (3,800 + 3,500 + 3,150) / 80 = 10,450 / 80$.
- Result.** **USD 130.63 per hour** (130.625 exactly; $\approx$ SAR 490 indicatively).
- Interpretation.** The blended rate makes options commensurable: a week of schedule compression staffed from this pool costs a computable amount, and shifting the mix changes the price as much as changing the headcount — adding five specialists moves the blend more than adding five technicians. This is the number that turns Domain 6's "crash it" instruction into an expected-value decision.

7.4.2 Contract models and cost risk

Every contract model is an answer to one question: **who carries cost risk?**

Model	Buyer pays	Cost risk sits with	Suits
Firm fixed price (FFP)	An agreed price, whatever it costs	Seller	Well-defined scope; priced-in risk premium
Fixed price incentive (FPIF)	Target cost/fee with a share ratio, capped by a ceiling	Shared, then seller above the ceiling	Definable scope with real uncertainty
Cost plus fixed fee (CPFF)	Allowable cost + a fixed fee	Buyer	Development, unclear scope
Cost plus incentive fee (CPIF)	Allowable cost + a fee varying with performance	Shared	Scope where effort is uncertain but effort quality matters
Time and materials (T&M)	Rates $\times$ quantities	Buyer	Staff augmentation, short or open-ended work

The leader's discipline: **risk transferred is risk priced**. An FFP contract for ill-defined scope does not remove the risk — it converts it into a premium plus a claims exposure when the scope moves (Domain 10, KA 10.4). Conversely a cost-plus contract on well-defined work pays for uncertainty that no longer exists.

7.4.3 Incentive fees and the point of total assumption

The mechanics. Under an incentive contract, buyer and seller agree a **target cost**, a **target fee** and a **share ratio** (buyer/seller) for over- and under-runs, with a **ceiling price** capping the buyer's exposure.

WORKED EXAMPLE

Worked example 7.4.3 — the incentive that stops incentivising.

1. **Setup.** Auriga's installation subcontract: target cost **USD 2,000,000**, target fee **USD 150,000**, share ratio **70/30** (buyer 70 %, seller 30 %), ceiling price **USD 2,450,000**. The seller finishes at an actual cost of **USD 2,300,000**.
2. **Formula.** Fee = target fee – (overrun × seller share). Buyer pays = actual cost + fee (subject to the ceiling). Point of total assumption $PTA = \text{target cost} + (\text{ceiling} - \text{target price}) / \text{buyer share}$, where target price = target cost + target fee.
3. **Substitution.** Overrun $2,300,000 - 2,000,000 = 300,000$; fee $150,000 - 300,000 \times 0.30 = 150,000 - 90,000$. Buyer pays $2,300,000 + 60,000$. Target price $2,150,000$; $PTA = 2,000,000 + (2,450,000 - 2,150,000) / 0.70$.
4. **Result.** Fee **USD 60,000** (down from 150,000); buyer pays **USD 2,360,000**. $PTA = \text{USD } 2,428,571.43$ — and at that cost the buyer pays exactly the USD 2,450,000 ceiling.
5. **Interpretation.** The share ratio does its job up to the PTA : both parties lose money on overrun, so both want efficiency. **Above the PTA the seller bears 100 %** of further cost — which is the moment the incentive inverts: a seller heading past it has no financial reason to spend more on your project and every reason to argue that the extra cost is your scope change. Knowing where the PTA sits is therefore a **delivery** insight, not an accounting one: it predicts when a commercial relationship will turn adversarial (Domain 10, KA 10.4; Domain 11's negotiation).

7.4.4 Cash flow versus profit

A project can be profitable and still fail for lack of cash — the same truth PFL-AI establishes for financings (its Domain 1). For a delivery leader the mechanism is payment terms: cost is incurred as work happens; cash arrives when invoices are approved and paid. On Auriga, with $AC = \text{USD } 2,120,000$ at week 13 and 60-day terms, roughly **USD 742,000** of incurred cost (about 35 %) is still unpaid — an exposure carried by whoever is funding the work. The leader's obligations are practical: know the terms in both directions (client and subcontractors), front-load nothing you cannot fund, and never let a retention or milestone-payment structure be agreed without someone computing its cash profile. Where the project sits inside a financed asset, this is precisely the CFADS conversation PFL-AI Domain 10 has with lenders.

AI in this KA

Contract analytics — extracting terms, comparing models, flagging inconsistent clauses — is real and useful AI assistance, and it is also decision support rather than legal or commercial advice. Two boundaries: an AI-produced reading of a clause is verified against the clause itself before anyone

relies on it (the document-against-summary check), and commercial and legal positions go to qualified counsel (Domain 10). Fee arithmetic and PTA computations are deterministic — they get the same golden-answer treatment as the earned-value set.

■ KEY TERMS — KA 7.4

Term	Meaning
Blended rate	Weighted average cost per hour of a mixed resource pool.
FFP / FPIF / CPFF / CPIF / T&M	Contract models ordered by who carries cost risk.
Share ratio	The agreed buyer/seller split of over- and under-run.
Ceiling price	The cap on the buyer's total exposure under an incentive contract.
Point of total assumption (PTA)	The cost above which the seller bears 100 % of further overrun.
Retention	A withheld percentage of payment, released on completion criteria.

■ SAMPLE MCQS — KA 7.4

MCQ 7.4-A [7.4.3 · Application] Target cost 2,000,000; target fee 150,000; share 70/30; actual cost 2,300,000. The seller's fee is: - A. USD 150,000 - B. USD 60,000 - C. USD 90,000 - D. USD 45,000

Rationale: The seller absorbs 30 % of the 300,000 overrun: $150,000 - 90,000 = 60,000$. A ignores the incentive; C states the fee reduction rather than the fee; D applies the buyer's share to the fee.

MCQ 7.4-B [7.4.3 · Analysis] Target cost 2,000,000, target fee 150,000, ceiling 2,450,000, buyer share 70 %. The PTA is 2,428,571, and its delivery significance is that above it: - A. the contract becomes void - B. the buyer absorbs all further cost - C. the seller bears 100 % of further cost, so the incentive inverts and cost growth becomes a scope-change argument - D. the fee becomes negative but risk-sharing continues unchanged

Rationale: Beyond the PTA the ceiling binds the buyer, so every further dollar is the seller's — which predictably redirects the seller's effort from efficiency to entitlement. B reverses the exposure; A is fiction; D misses that sharing has stopped.

MCQ 7.4-C [7.4.2 · Analysis] A leader lets an FFP contract for scope that is only 30 % defined. The most likely outcome is: - A. cost risk is genuinely eliminated - B. a priced-in risk premium plus a claims-and-variations exposure as the scope is defined - C. the seller absorbs all scope growth at no cost to the buyer - D. the contract converts automatically to cost-plus

Rationale: Fixed price transfers risk at a price and only for the scope actually specified; undefined scope returns as variations. A and C mistake the contractual form for the underlying uncertainty; D invents a mechanism.

MCQ 7.4-D [7.4.1 · Application] A pool of 40 at USD 95/h, 25 at USD 140/h and 15 at USD 210/h has a blended rate of: - A. USD 148.33 - B. USD 130.63 - C. USD 115.00 - D. USD 140.00

Rationale: $10,450/80 = 130.63$. A averages the three rates unweighted; C weights toward the cheapest grade only; D takes the middle rate as representative.

■ SELF-CHECK — KA 7.4

1. State the one question every contract model answers. — Who carries cost risk.

2. Why is the **PTA** a delivery concern, not just a commercial one? — Above it the seller bears all further cost, so behaviour shifts from efficiency to entitlement.
3. How can a profitable project run out of cash? — Cost is incurred as work happens; cash arrives on payment terms — the gap must be funded.

— Advanced topics — Domain 7

7.A.1 Earned schedule — closing **SPI**'s late-project blind spot

SPI converges on 1.00 as a project finishes, whatever its lateness, because **EV** and **PV** both approach **BAC**. **Earned schedule** restates progress in time: **ES** is the date at which the value now earned was planned to have been earned, and $SPI(t) = ES / AT$. Auriga's week-13 status, with value earned that the baseline expected by week 12, gives $SPI(t) = 12/13 = 0.92$ — and unlike **SPI** it stays meaningful to the last week. This is the bridge flagged in Domain 6 (KA 6.4.3), now with its cost-side companions.

7.A.2 EVM's limitations, stated plainly

Earned value measures conformance to plan, not value to the customer: a project can score 1.00 on both indices while building the wrong thing (Domain 5's acceptance criteria are the defence). It says nothing about quality (Domain 9). It is only as honest as its earning rules and its accruals. And it is blind to risk that has not yet materialised — Domain 8's contingency analysis, not **CPI**, tells you whether what remains is adequately funded. EVM is a measurement system, not a management philosophy; leaders who treat the indices as targets get optimised indices.

7.A.3 The reviewer's cost eye

Invariants a reviewer runs before trusting a cost report: **EV/BAC** equals the claimed percent complete; $CV = EV - AC$ and $SV = EV - PV$ reconcile to the stated indices; $EAC = AC +$ a stated **ETC**, with the assumption named; **TCPI** to **BAC/CPI** equals **CPI**; sum of control-account budgets + contingency equals **BAC**; management reserve sits outside that total; no completed package's budget has moved since last period; accruals are present in the current period; and level-of-effort share is disclosed per control account. Any violation is a defect somewhere — find it before the board builds on it.

— Industry variations — Domain 7

- **Construction and EPC.** Full EVM against a resource-loaded baseline; measured-work valuations and retention dominate cash; FIDIC-style variation and claims machinery makes the **PTA** conversation routine.
- **Government and defence programmes.** Formal EVMS compliance with surveillance, strict LOE caps and control-account discipline; forecasting is auditable rather than discretionary.
- **Technology and product delivery.** Cost is overwhelmingly people, so blended rates and capacity, not materials, drive the number; where cadence replaces a fixed baseline, throughput and cost-per-increment stand in for **CPI** (Domain 13), and hybrid programmes report both.

- **Energy and resources.** Long-lead equipment and commodity exposure put escalation and currency at the centre of the estimate (PFL-AI Domain 3's machinery); contingency is sized quantitatively as a matter of course.
- **Public services and transformation.** Benefits, not cost variance, are the accountability currency (Domain 16); the leader's task is keeping cost reporting honest while the value case is what gets debated.

— Case study — Domain 7: the forecast the board actually needed (utilities)

Situation. Auriga's week-13 review. The cost report shows [CPI](#) 0.91, [SPI](#) 0.92, [CV](#) (200,000), [SV](#) (160,000). The programme director's paper proposes reporting [EAC](#) **USD 4,200,000** — method (a) — on the grounds that the overrun was the contaminated-ground remediation, a discrete event now closed and remediated (Domain 6's case study).

The challenge. The assurance reviewer asks three questions. Is the cause genuinely closed? Yes for the remediation — but the recovery bought a second civil crew and a fast-track of installation, both of which continue for six more weeks, and neither is in method (a)'s "remaining work at budgeted rate" assumption. What does recovery to [BAC](#) require? [TCPI](#) 1.11 against a demonstrated 0.91. What is the mechanism? There isn't one — the recovery spend increases cost to protect the date.

The outcome. The board is given a **range with named assumptions**: 4,200,000 if the closed event is the whole story; 4,416,667 if current efficiency persists; and a bottom-up (method d) re-estimate of the remaining 52 %, commissioned that week, as the number they will actually manage to. Contingency draw is authorised against the identified risk; management reserve is not touched. The minute records the [TCPI](#) gap and the explicit finding that **no recovery-to-budget plan exists** — so nobody later claims one was promised.

What the domain teaches here. A single-number forecast hides the only thing that matters: the assumption. [TCPI](#) converts optimism into an arithmetic claim someone has to defend, and the honest report is the one that survives the question "what would have to be true?"

— Case study B — Domain 7: past the point of total assumption (technology)

Situation. A systems-integration subcontract ran on an incentive structure: target cost USD 2,000,000, target fee 150,000, 70/30 share, ceiling 2,450,000 — so [PTA](#) USD 2,428,571. By month eight the supplier's internal cost was tracking to 2,600,000, well past the [PTA](#).

What happened. The supplier's behaviour changed abruptly and, in hindsight, rationally: fresh change requests on work previously treated as in-scope, slower responses on defect fixes that earned nothing, and a claim that the buyer's late environment provision had caused the growth. The buyer's team read it as bad faith. It was arithmetic: past the [PTA](#) every additional engineer-hour came out of the supplier's own margin, and the only route back to profit was re-characterising cost as buyer-caused scope.

The outcome. The parties re-set commercially — a re-baselined target cost recognising the genuine environment delay (documented, and partly the buyer's), a revised ceiling, and a time-and-materials envelope for the disputed remainder. Delivery recovered within two months of the reset. The

retrospective's finding: the buyer had never computed the PTA, so a foreseeable inflection was experienced as a betrayal.

What the domain teaches here. Commercial structures create behaviour. Computing the PTA at signature — and watching the supplier's cost trend against it — turns an adversarial surprise into a managed conversation held early, while both parties still have options (Domain 10's supplier governance; Domain 11's negotiation).

— Executive perspective — Domain 7

What a project leader cannot delegate in this domain:

- **The forecast's assumption.** Analysts compute EAC; the leader owns which method and why, in one sentence a board can challenge. Reporting a number without its assumption is the domain's cardinal sin.
- **The TCPI reality test.** Before endorsing any recovery-to-budget plan: what index does the remaining work require, what is being demonstrated, and what specific mechanism closes the gap?
- **Reserve discipline.** Contingency and management reserve kept distinct, spent under stated authority, with consumption trended against progress — the erosion in MCQ 7.1-B is a leadership signal, not a bookkeeping detail.
- **Measurement integrity.** Earning rules fixed in advance, accruals present, level-of-effort share disclosed, no retrospective budget edits. Cost systems fail morally before they fail arithmetically — the same sentence as Domain 6, and the same reason.
- **The commercial shape.** Which model, why, and where the PTA sits. A leader who cannot say who carries cost risk on their largest package is not yet in control of it.

— Calculation exercises — Domain 7

Exercise 7.1 BAC 4,000,000; PV 2,080,000; EV 1,920,000; AC 2,120,000. Compute CV, SV, CPI, SPI, percent complete and percent spent. Solution. CV (200,000); SV (160,000); CPI 0.91; SPI 0.92; complete $1,920,000/4,000,000 = 48.0\%$; spent $2,120,000/4,000,000 = 53.0\%$. Common error: computing percent complete from AC/BAC — that is percent spent, and reporting it as progress overstates the project.

Exercise 7.2 Same data. Compute EAC by methods (a), (b) and (c), and the corresponding VAC. Solution. (a) $2,120,000 + 2,080,000 = 4,200,000$, VAC (200,000). (b) $4,000,000/0.905660 = 4,416,667$, VAC (416,667). (c) $2,120,000 + 2,080,000/(0.905660 \times 0.923077) = 4,608,056$, VAC (608,056). Common error: rounding CPI to 0.91 before dividing — $4,000,000/0.91 = 4,395,604$, USD 21,063 adrift. Indices are display; arithmetic is full precision.

Exercise 7.3 Same data. Compute TCPI to BAC and to the method-(b) EAC, and interpret. Solution. To BAC: $2,080,000/1,880,000 = 1.11$. To (b): $2,080,000/(4,416,667 - 2,120,000) = 2,080,000/2,296,667 = 0.91$. The second equals the current CPI — the identity of 7.3.4: BAC/CPI forecasts are "nothing changes". Common error: using PV rather than AC in the denominator (giving 1.08) and concluding recovery is nearly free.

Exercise 7.4 Incentive subcontract: target cost 2,400,000; target fee 180,000; share 80/20; ceiling 2,900,000. The seller finishes at 2,650,000. Find the fee, what the buyer pays, and the PTA. Solution. Overrun 250,000; fee $180,000 - 250,000 \times 0.20 = 130,000$; buyer pays $2,650,000 + 130,000 = 2,780,000$ (below the 2,900,000 ceiling). Target price 2,580,000; $PTA = 2,400,000 + (2,900,000 - 2,580,000)/0.80 = 2,400,000 + 400,000 = 2,800,000$. Common error: applying the buyer's 80 % share to the fee reduction.

Exercise 7.5 A pool of 30 at USD 105/h, 20 at USD 150/h and 10 at USD 220/h. Compute the blended rate, then the cost of adding four weeks of a 5-person crew at 40 h/week. Solution. $(30 \times 105 + 20 \times 150 + 10 \times 220)/60 = (3,150 + 3,000 + 2,200)/60 = 8,350/60 = \text{USD } 139.17/\text{h}$. Crew cost $5 \times 40 \times 4 \times 139.17 = \text{USD } 111,333$. Common error: using the blended rate for a crew drawn entirely from one grade — the blend applies to a representative mix, not to any arbitrary subset.

— Practitioner's toolkit — Domain 7

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 7.T.1 — Cost-report integrity checklist (run before publishing the month)

- [] Earning rules fixed in advance per package and unchanged this period; any change disclosed.
- [] EV claims sampled against physical evidence, not accepted on assertion.
- [] Accruals recognised for work received but not invoiced; open commitments cleansed.
- [] PV read from the current controlled baseline, phased over the approved schedule.
- [] No retro-fitted budgets; transfers between control accounts logged.
- [] EV/BAC reconciles to the reported percent complete; indices recompute from the four numbers.
- [] Every EAC states its method **and its assumption**; TCPI to BAC reported beside it.
- [] Contingency and management reserve shown separately, with consumption vs progress.
- [] Level-of-effort share disclosed per control account.
- [] AI-produced forecasts marked, calibration record attached, human owner named.

Toolkit 7.T.2 — Estimate basis sheet (one per estimate)

Method (analogous/parametric/bottom-up) · definition maturity and accuracy class · range (–x %/+y %) · rate sources with dates and escalation basis · quantities and their source · exclusions stated explicitly · risks feeding contingency (link to the register) · reviewer and date. An estimate whose basis sheet cannot be produced is not releasable.

Toolkit 7.T.3 — Commercial one-pager (per major package)

Model (FFP/FPIF/CPFF/CPIF/T&M) and why · who carries cost risk · target cost, fee, share ratio, ceiling · **PTA and current cost trend against it** · payment terms both directions and the cash profile · retention and release criteria · variation and claims route · escalation contacts. Reviewed at every commercial checkpoint, not filed at signature.

— Exam preparation — Domain 7

The calculation traps. Percent spent reported as percent complete (Exercise 7.1) · rounding **CPI** before dividing (Exercise 7.2) · **PV** instead of **AC** in the **TCPI** denominator (Exercise 7.3) · applying the wrong party's share to an incentive fee (Exercise 7.4) · **EV** valued at actual cost rather than budget · reading a level-of-effort-heavy **SPI** as schedule performance (MCQ 7.3-D) · quoting **SPI** late in a project where it must converge on 1.00 (7.A.1) · confusing commitments with actuals · treating contingency and management reserve as one pot.

Reflection questions. 1. Your cost report shows one **EAC**. What must accompany it before you will sign it? (The method and the assumption about remaining work; **TCPI** to **BAC** beside it.) 2. On your largest package: who carries cost risk, where is the **PTA**, and how close is the supplier's cost trend to it? (7.4.2–7.4.3; toolkit 7.T.3.) 3. Which invariant in 7.A.3 would have caught the last cost surprise you experienced — and why wasn't it running?

— Domain 7 summary

Cost leadership begins with an honest number: a method matched to definition maturity, a range and class always stated, three-point thinking where tails are real, and a budget built through control accounts into a time-phased baseline with contingency inside it and management reserve outside. Measurement then has to be earned — **AC** with accruals, **EV** at budget under earning rules fixed in advance — because everything downstream is arithmetic on those numbers: **CV**, **SV**, **CPI**, **SPI** at Auriga's week 13 (0.91 and 0.92; 48 % complete against 53 % spent), the **EAC** family spanning USD 408,056 on identical data because they encode different assumptions, and **VAC** and **TCPI** turning recovery talk into a defensible index (1.11 required against 0.91 demonstrated). The commercial half is the same discipline pointed outward: blended rates make options commensurable, contract models allocate cost risk at a price, and the point of total assumption predicts when a supplier's incentives invert. Throughout, the leadership rule holds — the number is only as good as the assumption named beside it, and machine-produced forecasts reach a board only with a calibration record and a human owner. Domain 8 quantifies the risk this domain reserves for; Domain 10 takes the commercial relationships into their own depth.

08

DOMAIN 8

Risk, Uncertainty and Resilience (quantitative)

Group: Delivering the work (Part Two). **Target:** ~74 pages. **Binds to:** the PCI Book Pattern Specification and the shared registries ([docs/books/registries/](#)). This domain is the home of **EMV** and supplies the quantification behind the contingency that Domain 7 (KA 7.1.3) reserves and the schedule ranges Domain 6 (KA 6.4.3) forecasts with. British English; USD (+SAR where useful, indicative **USD 1 ≈ SAR 3.75**).

— Why this domain exists

Domains 6 and 7 built plans and budgets; both left an obligation outstanding. Domain 6 forecast with ranges but did not say where the ranges come from. Domain 7 put a contingency reserve inside the baseline and said it is "sized against identified risks" without showing how. This domain closes both gaps and adds the part neither covers: what a leader does when the unexpected happens anyway. It builds risk identification honestly, including opportunities (KA 8.1); works the analysis from qualitative screening through **EMV**, decision trees and aggregated contingency (KA 8.2); sizes and governs reserves (KA 8.3); and turns to resilience, crisis leadership, cognitive bias and AI-enabled risk sensing (KA 8.4). The discipline throughout is that **risk management is a decision-support activity, not a documentation activity** — a register that changes no decision has cost money and bought nothing.

Learning objectives. After this domain a candidate can: distinguish risk from uncertainty and issue; write a risk statement that supports a decision; identify opportunities as rigorously as threats; run qualitative screening and state its limits; compute **EMV** for individual risks and interpret the total; build and solve a decision tree, including the value of information; aggregate independent risks to a mean and a confidence level rather than summing point estimates; size a contingency reserve defensibly and distinguish it from management reserve; select and cost responses; explain resilience as distinct from prediction; recognise the biases that corrupt estimates and reviews; lead in a crisis; and govern AI-produced risk analysis.

The master worked project. Project Auriga continues from Domains 6 and 7 — the 25-week, **BAC USD 4,000,000** control-systems upgrade. Its risk register, used throughout KA 8.2–8.3, carries the ground-conditions risk that actually materialised in Domain 6's case study, so the reader sees the same event before and after the fact.

KNOWLEDGE AREA 8.1

Threats, opportunities and identification

Topics: 8.1.1 risk, uncertainty and issues · 8.1.2 writing a usable risk statement · 8.1.3 identification methods and their blind spots.

8.1.1 Risk, uncertainty and issues

Definitions. A **risk** is an uncertain event that, if it occurs, affects an objective — threats reduce achievement, **opportunities** improve it. **Uncertainty** in the broader sense covers what cannot be enumerated with probabilities at all — the conditions where scenario thinking and resilience substitute for calculation. An **issue** is a risk that has occurred; it is managed, not analysed. Keeping these apart matters practically: registers clogged with issues stop being forward-looking, and treating deep uncertainty as if it were quantifiable produces false precision (KA 8.4.1).

Risk appetite and thresholds convert judgment into something operable: how much exposure the organisation accepts, expressed as thresholds that trigger escalation or response — connecting this domain to Domain 3's decision rights. Appetite that is never expressed numerically cannot be applied consistently by different people, which is the usual reason two projects in one portfolio treat the same exposure differently.

8.1.2 Writing a usable risk statement

Most registers fail at the sentence level. A usable statement has three parts — **cause** → **event** → **consequence** — and names the affected objective:

Because the controller is single-sourced with a volatile lead time (**cause**), it may be that delivery slips beyond the installation window (**event**), resulting in a delay to commissioning and additional preservation cost (**consequence**).

Compare "supplier risk", which supports no decision: it identifies no cause to attack, no event to monitor and no consequence to size. The three-part form is not a formatting preference — each part maps to a different response type (attack the cause, monitor the event, mitigate the consequence), which is why KA 8.3's response selection depends on it.

8.1.3 Identification methods and their blind spots

Methods: structured workshops, checklists and prompt lists from prior projects, assumption analysis (every assumption is a risk in disguise — Domain 2's assumption register), interviews for what people will not say in a group, and lessons from comparable work (Domain 9's lessons-learned). Each has a blind spot, and knowing them is the professional part:

- **Workshops** surface what the group already shares and suppress what a junior member suspects — so run them with an explicit invitation to dissent and a route for private input.
- **Checklists** find known risks and, by their comfort, discourage looking for novel ones.
- **Assumption analysis** is the highest-yield method and the least used, because it requires admitting how much of the plan is assumed.
- **Historical data** encodes the past's risks, not this project's novelties.

Opportunities are systematically under-identified — most registers are 90 % threats — because the process is framed defensively and because nobody is rewarded for one. A leader who wants them must ask for them separately and size them the same way (Auriga's R5 below is one).

AI in this KA

Generating candidate risk lists is a real AI strength: broad, fast, and free of the group's shared blind spot — genuinely useful against the workshop failure mode above. It is equally a machine for **plausible generic risk**, producing items that fit any project and support no decision. The governed workflow: use AI to widen identification (especially by asking it to challenge the plan's assumptions), then require every surviving item to pass the three-part statement test of 8.1.2 with a named owner. **AI proposes; the professional verifies, decides and remains accountable** — and here the verification is specifically that the risk is this project's, not a template's.

■ KEY TERMS — KA 8.1

Term	Meaning
Risk / opportunity	Uncertain event affecting an objective, adversely / favourably.
Uncertainty	Conditions that cannot be meaningfully enumerated with probabilities.
Issue	A risk that has occurred; managed, not analysed.
Risk appetite / threshold	Accepted exposure, expressed operably enough to apply consistently.
Cause → event → consequence	The three-part risk statement; each part maps to a response type.

■ SAMPLE MCQS — KA 8.1

MCQ 8.1-A [8.1.2 · Analysis] Which register entry best supports a decision? - A. "Supplier risk — high" - B. "Because the controller is single-sourced with volatile lead times, delivery may slip beyond the installation window, delaying commissioning and adding preservation cost" - C. "Delay to commissioning — probability 35 %" - D. "Supplier may cause problems — owner: procurement"

Rationale: Only B names a cause to attack, an event to monitor and a consequence to size (8.1.2). C states a consequence with a probability but no cause, so no response can be designed; A and D support nothing at all.

MCQ 8.1-B [8.1.1 · Recall] The ground contamination has been discovered and remediation is under way. In the register this is: - A. a risk with probability 1.0 - B. an issue — it has occurred, and is managed rather than analysed - C. an opportunity, since remediation was funded - D. removed entirely, with no further record

Rationale: Occurred risks become issues (8.1.1). A is the common fudge that clogs registers; D loses the audit trail and the lesson; C is nonsense.

MCQ 8.1-C [8.1.3 · Analysis] A register of 60 items contains 3 opportunities. The soundest inference is: - A. this project genuinely has few opportunities - B. the identification process is framed defensively; opportunities must be asked for separately and sized the same way - C. opportunities do not belong in a risk register - D. the register is too long and should be cut

Rationale: A 95 % threat ratio reflects process framing rather than reality (8.1.3). C contradicts the definition of risk; D addresses a different problem.

■ SELF-CHECK — KA 8.1

1. State the three parts of a usable risk statement. — Cause, event, consequence, against a named objective.
2. Why is assumption analysis the highest-yield identification method? — Every assumption is a risk in disguise, and plans rest on more of them than teams admit.
3. What is the register consequence of treating an occurred risk as a 100 % risk? — It clogs the forward-looking register; occurred risks are issues.

KNOWLEDGE AREA 8.2

Analysis: from screening to quantification

Topics: 8.2.1 qualitative screening and its limits · 8.2.2 expected monetary value · 8.2.3 decision trees and the value of information · 8.2.4 aggregating to a contingency.

8.2.1 Qualitative screening and its limits

Probability-and-impact scoring on a matrix is a **screening** tool: cheap, fast, and adequate for ranking a long list into an attention order. Its limits are structural and must be stated wherever it is used. Ordinal scales cannot be arithmetic — a "4" is not twice a "2", so scores must not be multiplied and summed as if they were money. Band boundaries create false distinctions (a risk at the top of "medium" outranks one at the bottom of "high" in reality but not on the matrix). And the matrix is blind to **correlation**: five risks driven by one supplier are one risk wearing five costumes.

The professional practice: screen qualitatively, then **quantify what matters** — the items that could breach a threshold, plus anything a decision now depends on.

8.2.2 Expected monetary value

$EMV = \text{probability} \times \text{impact}$ (per risk; threats positive, opportunities negative cost)

WORKED EXAMPLE

Worked example 8.2.2 — Auriga's risk register, quantified.

1. **Setup.** Four threats and one opportunity, each with an assessed probability and cost impact (BAC USD 4,000,000).
2. **Formula.** $EMV = p \times \text{impact}$; total exposure = ΣEMV , opportunities carried negative.
3. **Substitution and result.**

ID	Risk	p	Impact (USD)	EMV (USD)
R1	Controller lead-time slip	0.35	240,000	84,000
R2	Ground conditions worse than surveyed	0.50	180,000	90,000
R3	Integration rework	0.25	320,000	80,000
R4	Permit delay	0.15	400,000	60,000
R5	Early-delivery rebate (opportunity)	0.30	(120,000)	(36,000)
Total expected exposure				278,000

1. **Interpretation.** Total EMV is **USD 278,000** — **6.95 %** of BAC, and USD 314,000 if the opportunity is ignored, which is why opportunities must be in the same arithmetic rather than mentioned in prose. Two cautions the table itself teaches. **EMV is an average of outcomes that will not happen:** no single risk will cost 84,000 — R1 costs 240,000 or nothing. So EMV is the right basis for funding a portfolio of risks and the wrong basis for deciding whether one specific risk is survivable. And **ranking by EMV reorders the register:** R4 has the largest impact (400,000) but the smallest threat EMV (60,000), while R3's lower impact carries more expected cost. A leader managing by impact alone would attend to them in the wrong order — though see 8.3.1, because impact still governs survivability.

8.2.3 Decision trees and the value of information

A decision tree makes a choice under uncertainty explicit: decision nodes (what we control), chance nodes (what we do not), and outcomes valued and rolled back to the present decision.

WORKED EXAMPLE

Worked example 8.2.3 — should Auriga survey before excavating?

1. **Setup.** Excavation may hit contamination: probability **0.40**, cost if hit **USD 300,000**. Option A: proceed directly. Option B: commission a **USD 25,000** ground survey first, which reliably reveals the condition; if contamination is present, a planned mitigation costs **USD 90,000** instead of the reactive 300,000.
2. **Formula.** Roll back each branch: $EV(\text{cost}) = \sum \text{probability} \times \text{outcome cost}$. Value of information = $EV(\text{cost without}) - EV(\text{cost with})$.
3. **Substitution.** A: $0.40 \times 300,000$. B: $25,000 + 0.40 \times 90,000 = 25,000 + 36,000$.
4. **Result.** A **USD 120,000** expected cost; B **USD 61,000**. **Value of the survey = USD 59,000**, against its USD 25,000 price.
5. **Interpretation.** The survey is worth buying, and the arithmetic shows why: it converts a reactive 300,000 into a planned 90,000 in the 40 % of futures where the problem exists. This is the general shape of **buying information** — worth it when it changes what you would do, worth nothing when it does not, however reassuring it feels. Note the sensitivity: if mitigation once forewarned still cost 250,000, B's EV would be $25,000 + 100,000 = 125,000$ and the survey would destroy value. **Information is valuable only in proportion to the better action it enables.** (Domain 6's case study is the counterfactual: Auriga did not survey, met the condition reactively, and paid the recovery this tree prices.)

Survey worth buying: it converts a reactive 300,000 into a planned 90,000 in the 40 % of futures where the problem exists

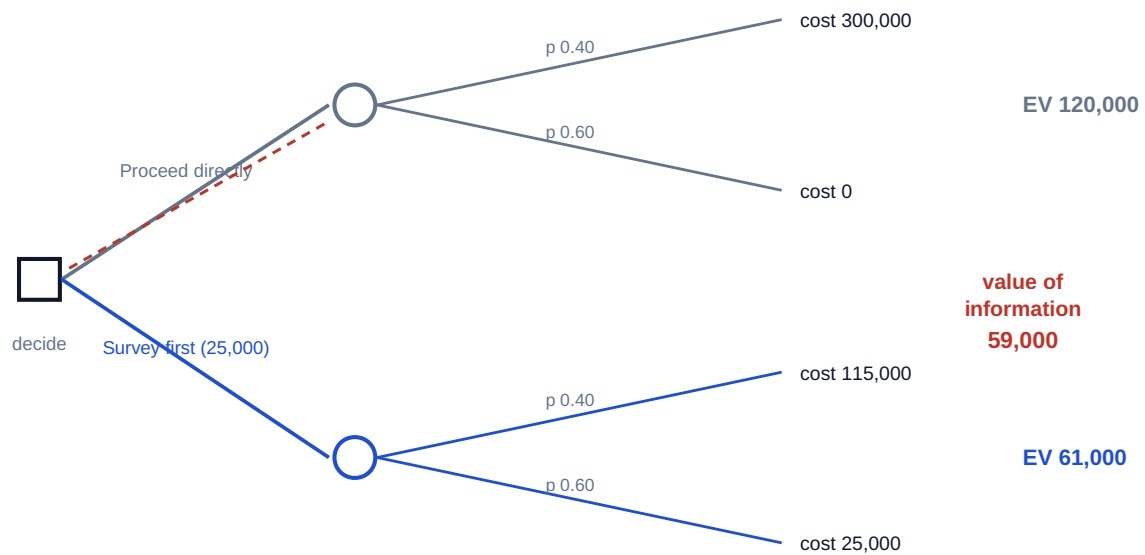


Fig 8.2.1 — The survey decision tree

8.2.4 Aggregating to a contingency

The error to avoid. Summing worst cases produces a number nobody will fund ($240,000 + 180,000 + 320,000 + 400,000 = 1,140,000$, or 28.5 % of BAC); summing EMVs alone (278,000) funds the average, which by construction is exceeded about half the time. Neither is a contingency. What a leader needs is a **confidence level**: an amount that covers the aggregate outcome with stated probability.

WORKED EXAMPLE**Worked example 8.2.4 — Auriga's contingency at P80.**

1. **Setup.** The five register risks of 8.2.2, treated as independent (an assumption examined below).
2. **Formula.** For independent risks: mean = $\sum p \times \text{impact}$; variance = $\sum p(1-p) \times \text{impact}^2$; standard deviation = $\sqrt{\text{variance}}$. A P80 amount $\approx \text{mean} + 0.8416 \times \sigma$ (the 80th percentile of a normal approximation).
3. **Substitution.** Mean **278,000** (from 8.2.2). Variance $0.35 \times 0.65 \times 240,000^2 + 0.50 \times 0.50 \times 180,000^2 + 0.25 \times 0.75 \times 320,000^2 + 0.15 \times 0.85 \times 400,000^2 + 0.30 \times 0.70 \times 120,000^2 = \mathbf{63,828,000,000}$; $\sigma = \mathbf{252,642}$. P80 = $278,000 + 0.8416 \times 252,642$.
4. **Result.** σ **USD 252,642**; **P80 $\approx$ USD 490,624**. Against the worst-case sum of 1,140,000 and the EMV sum of 278,000 — and against the "10 % of BAC" rule of thumb, which would give 400,000 with no reasoning attached.
5. **Interpretation.** The P80 figure is defensible in a way the other three are not: it states what confidence it buys. Three professional caveats belong beside it. **Independence is an assumption, usually optimistic** — if R1 and R3 share a supplier they are correlated, the variance is larger, and the true P80 is higher; correlation is what turns a bad month into a crisis. **The normal approximation is a convenience** for a handful of Bernoulli risks; proper practice runs a Monte Carlo simulation over the register (and over the schedule, extending Domain 6's three-point durations) to produce a distribution rather than a formula. And **the confidence level is a policy choice**, not a technical one — P80 is common for contingency, P50 for a target, and whoever chooses is making a risk-appetite decision (8.1.1) that belongs to governance.

AI in this KA

Simulation, correlation analysis and scenario generation are legitimate machine work, and the outputs are unusually seductive because a distribution looks like evidence. Three checks before any simulated number reaches a board. **Interrogate the inputs:** a simulation is an amplifier of its assumed distributions and correlations, and those are judgments — garbage in, precisely-shaped garbage out. **Recompute a landmark by hand:** the mean should match $\sum \text{EMV}$; if it does not, the model is not doing what you think. **Ask what would change the decision:** a P80 quoted to the dollar with no sensitivity is decoration. The register's inputs remain human judgments with named owners, whatever produced the output.

■ KEY TERMS — KA 8.2

Term	Meaning
EMV	probability $\times$ impact; the average of outcomes that will not individually occur.
Decision tree	Explicit rollback of decisions and chance events to a present choice.
Value of information	The reduction in expected cost that knowing enables; zero if it changes no action.
P50 / P80	Confidence levels — amounts covering the aggregate outcome with stated probability.
Correlation	Shared drivers that raise aggregate variance; independence is usually optimistic.
Monte Carlo simulation	Repeated sampling to produce an outcome distribution rather than a point.

■ SAMPLE MCQS — KA 8.2

MCQ 8.2-A [8.2.2 · Application] A risk has probability 0.15 and impact USD 400,000. Its **EMV** is: - A. USD 400,000 - B. USD 60,000 - C. USD 340,000 - D. USD 26,667

Rationale: $0.15 \times 400,000 = 60,000$. A is the impact; C is impact less **EMV**; D divides instead of multiplying.

MCQ 8.2-B [8.2.2 · Analysis] R4 has the register's largest impact (400,000) but its smallest threat **EMV** (60,000). The correct managerial reading is: - A. R4 should be ignored, having the lowest **EMV** - B. **EMV** sets the funding priority, while impact still governs whether the event is survivable — both readings are needed - C. impact alone should drive priority - D. the assessment must be wrong, since impact and **EMV** disagree

Rationale: **EMV** is the right basis for funding a portfolio of risks; a single large impact may still be existential regardless of probability (8.2.2, 8.3.1). A and C each discard half the information; D misunderstands that the two measure different things.

MCQ 8.2-C [8.2.3 · Application] Proceeding directly has an expected cost of $0.40 \times 300,000$. A USD 25,000 survey reduces the mitigated cost to USD 90,000. The value of the information is: - A. USD 25,000 - B. USD 59,000 - C. USD 84,000 - D. nil — the survey costs more than it saves

Rationale: $120,000 - (25,000 + 36,000) = 59,000$. A is the survey's price, not its value; C omits the survey cost from branch B; D reverses the conclusion.

MCQ 8.2-D [8.2.4 · Analysis] A register's **EMV** sum is 278,000 and its worst-case sum is 1,140,000. Setting contingency at 278,000 means: - A. an appropriately funded reserve - B. funding the average outcome, which by construction is exceeded roughly half the time - C. a conservative reserve, since not all risks will occur - D. the same as a P80 reserve

Rationale: The mean is the ~50th percentile of the aggregate (8.2.4), so it is exceeded about half the time — the reason a confidence level is chosen explicitly. C mistakes the mean for conservatism; D confuses two different statistics (490,624 here).

MCQ 8.2-E [8.2.1 · Analysis] Why must ordinal probability-impact scores not be multiplied and summed as money? - A. the matrix is only for threats - B. ordinal bands are ranks, not quantities — a "4" is not twice a "2", so the arithmetic is meaningless - C. multiplication requires more than five bands - D. scores may be summed provided they are weighted

Rationale: Ordinal scales support ordering, not arithmetic (8.2.1). Weighting (D) does not repair a scale that never carried magnitude.

■ SELF-CHECK — KA 8.2

1. Why is **EMV** wrong for judging whether one risk is survivable? — It averages outcomes that will not occur; the actual event is impact-or-nothing.
2. When is information worth nothing however reassuring? — When it would not change the action taken.
3. Which assumption in the P80 calculation is usually optimistic, and why does it matter? — Independence; correlated risks raise variance and the true confidence amount.

KNOWLEDGE AREA 8.3

Responses, reserves and governance

Topics: 8.3.1 selecting and costing responses · 8.3.2 reserves and their authority · 8.3.3 monitoring and the register that earns its keep.

8.3.1 Selecting and costing responses

The response families — **avoid** (remove the cause or the exposure), **reduce** (lower probability or impact), **transfer** (insurance, contract terms — a price, not a disappearance; Domain 7, KA 7.4.2), **accept** (with or without a fallback), and for opportunities **exploit, enhance, share, ignore**. Two rules make selection professional rather than reflexive.

A response is an investment. It costs money and time and reduces **EMV**; if the reduction is smaller than the cost, the response destroys value and acceptance is the correct answer. Auriga's fast-track decision from Domain 6 is exactly this arithmetic: one week of client bonus (USD 45,000) against an accepted rework risk of $0.20 \times 60,000 = 12,000$ — a net **+USD 33,000**, which is why the recovery plan took it.

Impact governs survivability even at low probability. A 3 % chance of an event the project cannot survive is not managed by its **EMV**; it is avoided, transferred, or the project's viability is reconsidered. **EMV** funds portfolios; existential risks are handled by structure, and confusing the two is how organisations optimise their way into single points of failure.

Secondary risks. Every response creates its own — fast-tracking creates rework risk, a transfer creates counterparty risk, mitigation creates delivery risk on the mitigation itself. A register whose responses have no secondary entries has not been thought through.

8.3.2 Reserves and their authority

Domain 7 (KA 7.1.3) established the structure; this domain supplies the sizing and the governance:

Reserve	Covers	Sized by	Spent by
Contingency	Identified risks in the register	Aggregation to a stated confidence (8.2.4)	Project manager, under a published protocol
Management reserve	Unknown-unknowns and scope change	Judgment and organisational policy	Sponsor / change authority, via change control

Three governance rules. **The draw protocol is published in advance** — which risk, what evidence, what approval — because a reserve released ad hoc is indistinguishable from an overrun. **Consumption is trended against risk retirement**, not against time: burning 60 % of contingency while 20 % of the register has been retired is the signal MCQ 7.1-B describes. And **contingency released by retired risks is returned, not reallocated** to convenient overspends; otherwise the reserve silently becomes a slush fund and the next real risk is unfunded.

8.3.3 Monitoring and the register that earns its keep

A live register has movement: probabilities revised as evidence arrives, risks retired when their window closes, new entries as the project learns, owners who report rather than merely appear. The tests of whether it is working are behavioural, not documentary — **has a decision changed because of it this month?, are the top items the ones the team actually worries about?**, and

did anything that hurt us appear in it beforehand? A register that answers no to all three is theatre, and the honest response is to fix the process rather than reformat the document. **Early-warning indicators** are the register's operational edge: leading measures tied to specific risks (supplier confirmations slipping, defect discovery rate rising, permit queue lengthening) that fire before the event, which is what makes a response affordable (Domain 6's lead-time point).

■ KEY TERMS — KA 8.3

Term	Meaning
Avoid / reduce / transfer / accept	The threat-response families; transfer is a price, not a disappearance.
Secondary risk	Risk created by a response; a register without them is incomplete.
Draw protocol	The published rules for releasing contingency.
Risk retirement	Closing a risk whose window has passed; frees contingency for return.
Early-warning indicator	A leading measure tied to a risk, firing before the event.

■ SAMPLE MCQS — KA 8.3

MCQ 8.3-A [8.3.1 · Application] A mitigation costs USD 50,000 and reduces a risk's **EMV** from 84,000 to 20,000. The decision and its basis are: - A. reject — 50,000 is a large outlay - B. accept — the **EMV** reduction of 64,000 exceeds the 50,000 cost - C. accept — any reduction in **EMV** justifies a response - D. indifferent — cost and benefit are equal

Rationale: Responses are investments: $84,000 - 20,000 = 64,000$ of reduction for 50,000 of cost is value-creating. A prices the outlay without the benefit; C would justify unlimited spend; D miscomputes.

MCQ 8.3-B [8.3.1 · Analysis] A risk has probability 0.03 and an impact the project could not survive. The correct treatment is: - A. accept it — the **EMV** is small - B. treat it structurally: avoid, transfer, or reconsider viability, because **EMV** funds portfolios while survivability is governed by impact - C. fund its **EMV** in contingency and monitor - D. exclude it from the register as improbable

Rationale: Existential exposure is not an averaging problem (8.3.1). A and C both apply portfolio logic to a single point of failure; D removes the entry that most needs governance attention.

MCQ 8.3-C [8.3.2 · Analysis] Contingency freed by a retired risk is used to cover an unrelated overspend. This is: - A. efficient reserve management - B. a governance failure: the reserve silently becomes a slush fund and the next genuine risk is unfunded - C. acceptable if the total baseline is unchanged - D. required, since contingency is inside the baseline

Rationale: Contingency is tied to identified risks; reallocating it to overspends destroys the link between reserve and exposure (8.3.2). C is the reasoning that makes the failure invisible.

■ SELF-CHECK — KA 8.3

1. When is acceptance the professionally correct response? — When the response's cost exceeds the **EMV** reduction it buys — and the risk is survivable.
2. What does transfer actually achieve? — It prices the risk to a counterparty and creates counterparty risk; it does not remove the exposure.
3. What should contingency consumption be trended against? — Risk retirement, not elapsed time.

KNOWLEDGE AREA 8.4

Resilience, bias and crisis leadership

Topics: 8.4.1 resilience versus prediction · 8.4.2 cognitive bias · 8.4.3 crisis leadership · 8.4.4 AI-enabled risk sensing.

8.4.1 Resilience versus prediction

Quantification (KA 8.2) handles risks you can name. **Resilience** is the capability to absorb what you cannot: it assumes the register is incomplete, because it always is. Its levers are structural rather than analytical — **buffers** (float placed deliberately, contingency, capacity headroom), **optionality** (staged commitments, alternative suppliers qualified in advance, reversible decisions — Domain 1, KA 1.3.3), **modularity** (a failure contained rather than propagated across interfaces), **redundancy** where failure is intolerable, and **fast detection** (early-warning indicators, short feedback loops), because response cost rises with delay.

The trade-off is real and should be stated rather than finessed: resilience costs efficiency. A project optimised to the last dollar and week has no capacity to absorb surprise, and one buffered everywhere is uncompetitive. The leadership judgment is where to be resilient — at the points where failure propagates or cannot be tolerated — not whether.

8.4.2 Cognitive bias

Estimates and reviews are produced by people, and the predictable distortions are worth naming because each has a countermeasure:

Bias	Effect	Countermeasure
Optimism bias / planning fallacy	Systematic underestimation of duration and cost	Reference-class forecasting against comparable completed projects
Anchoring	First number dominates subsequent estimates	Independent estimates before any figure is shared
Availability	Recent or vivid events overweighted	Structured checklists and historical base rates
Groupthink / social pressure	Dissent suppressed in workshops	Pre-mortem, private input channels, explicit dissent invitation
Sunk-cost / escalation of commitment	Continuing because of what is spent	Decision framed on remaining cost and benefit only (Domain 2's kill criteria)
Confirmation bias	Evidence sought that supports the plan	Assign someone to argue the opposite; test the disconfirming case

The **pre-mortem** deserves particular mention as the highest-yield, cheapest technique in this domain: before committing, ask the team to imagine the project has failed and to write down why. It licenses the dissent a status meeting suppresses, and it reliably surfaces risks the workshop of 8.1.3 missed.

8.4.3 Crisis leadership

When a serious risk materialises, the leader's job changes shape. The sequence that works: **stabilise** (stop the harm; secure safety, then the situation), **establish facts** (separate what is known from

assumed — crises run on rumour), **decide with a clock** (a good decision now beats an optimal one later, and say explicitly which decisions are reversible), **communicate early and honestly** (bad news does not improve with age; Domain 11's stakeholder work is now executing under compressed time), **protect the team** (crises consume people, and exhausted teams make the second mistake), and **capture the lesson while it is vivid** (Domain 9's lessons-learned). Domain 6's recovery machinery is the schedule expression of the same posture: re-run the passes, price the options, escalate the trade with named authority.

8.4.4 AI-enabled risk sensing

The genuine capability is **detection at scale**: anomaly detection across cost, schedule, quality and supplier data; pattern recognition against historical projects; and sentiment or volume signals in issue logs and correspondence that precede formal escalation. Used well, this shortens the detection half of 8.4.1's fast-detection lever — which is where its value actually lies.

The honest boundaries: models detect **patterns like the past**, so novel risk is invisible to them; they produce **false positives** that erode attention if unfiltered; and a **calibration record is mandatory** before their alerts influence decisions (Domain 6, KA 6.4; Domain 14). The failure mode to watch for is not a wrong alert but **displaced judgment** — a team that stops thinking about risk because a dashboard is watching. The register's owners remain human, the analysis stays challengeable, and the principle holds: AI proposes; the professional verifies, decides and remains accountable.

■ KEY TERMS — KA 8.4

Term	Meaning
Resilience	Capability to absorb unidentified risk; buffers, optionality, modularity, redundancy, fast detection.
Reference-class forecasting	Estimating from comparable completed projects to counter optimism bias.
Pre-mortem	Imagining failure before committing, to license dissent and surface missed risk.
Escalation of commitment	Continuing because of sunk cost; countered by forward-looking framing.
Displaced judgment	Ceasing to think because a tool is watching.

■ SAMPLE MCQS — KA 8.4

MCQ 8.4-A [8.4.2 · Application] A programme's estimates have run 25 % low across four comparable past projects. The most effective countermeasure is: - A. instruct estimators to be more careful - B. reference-class forecasting — estimate from the distribution of comparable completed projects rather than from the plan - C. add a 25 % contingency and continue the same process - D. escalate to the sponsor for a larger budget

Rationale: Optimism bias is systematic, so the fix is methodological (8.4.2). A relies on exhortation against a structural effect; C treats the symptom while preserving the cause; D funds it without correcting it.

MCQ 8.4-B [8.4.1 · Analysis] A project has been optimised to eliminate all float and buffers. Its risk position is: - A. improved — waste has been removed - B. degraded — it now has no capacity to absorb the risks not in the register, and resilience is what covers those - C. unchanged, provided contingency is funded - D. improved, since efficiency reduces exposure time

Rationale: Resilience assumes register incompleteness (8.4.1); removing all absorption capacity maximises fragility. C confuses money with time and capacity — funded contingency cannot buy back a schedule with nowhere to move.

MCQ 8.4-C [8.4.4 · Analysis] An AI monitor has raised 40 alerts this month, 3 of which mattered. The correct response is: - A. disable the monitor - B. tune thresholds and require a calibration record, because unfiltered false positives erode the attention the tool exists to direct - C. investigate all 40 equally - D. accept the ratio as inherent to anomaly detection

Rationale: False positives consume the scarce resource (attention) that detection is meant to focus (8.4.4). A discards real capability, C guarantees the erosion, D abandons the calibration duty.

MCQ 8.4-D [8.4.3 · Recall] In the first hour of a crisis, the leader's priority order is: - A. communicate, then investigate, then stabilise - B. stabilise and secure safety, establish facts, then decide with a clock and communicate early - C. establish blame, then stabilise - D. wait for complete information before acting

Rationale: Stabilisation and factual grounding precede decisions, which are taken against a clock (8.4.3). D is the failure the sequence exists to prevent; C is never a first-hour activity.

■ SELF-CHECK — KA 8.4

1. Why is resilience not a substitute for quantification, and vice versa? — Quantification handles named risks; resilience absorbs the unnamed. Each fails at the other's job.
2. What is a pre-mortem for? — To license dissent before commitment and surface risks a workshop suppresses.
3. What is AI risk sensing's structural blind spot? — Novel risk: it detects patterns like the past.

— Advanced topics — Domain 8

8.A.1 Correlation and why aggregate risk is worse than it looks

Independence made 8.2.4's arithmetic tractable and understated the answer. When risks share a driver — one supplier, one technology, one regulator, one labour market — they move together, and correlated variance is additive in a way that widens the aggregate distribution's tail sharply. The practical implications: identify **common drivers** explicitly and group risks by them; model correlation in simulation rather than assuming it away; and treat a register with many entries and few drivers as concentrated, not diversified. The organisational version of this error is believing a portfolio of projects is diversified when every one depends on the same scarce engineering resource.

8.A.2 Schedule risk analysis and the merge-bias trap

Applying this domain's methods to Domain 6's network produces **quantitative schedule risk analysis**: three-point durations (KA 6.4.3) simulated across the logic to yield a completion distribution and, valuably, a **criticality index** — how often each activity lands on the critical path across simulations — which is more decision-useful than a single deterministic critical path. The trap it exposes is **merge bias**: where paths converge (Domain 6's node E), the merged event waits for the latest predecessor, so the probability of being on time is the product of the paths' probabilities, not the best of them. Two paths each 80 % likely to meet a date give the merge point 64 %. Deterministic CPM cannot see this, which is why convergence points are systematically optimistic and why Domain 6 flagged them for attention.

8.A.3 The reviewer's risk eye

Invariants worth testing in an hour: every entry has cause, event and consequence and a named owner; no occurred risks parked as high-probability entries; opportunities present in credible proportion; Σ EMV reconciles to the register; contingency stated at a named confidence level with its independence assumption disclosed; contingency consumption trended against risk retirement, not time; responses costed against EMV reduction, with secondary risks recorded; existential risks handled structurally rather than by EMV; early-warning indicators defined for the top items; and at least one decision this month traceable to the register. Failure of the last one matters most: it means the process is documentation, not decision support.

— Industry variations — Domain 8

- **Construction and infrastructure.** Ground conditions, weather and permits dominate; quantitative schedule risk analysis is contractually expected on major programmes, and risk allocation is largely executed through contract terms (Domain 10).
- **Energy and resources.** Commodity price and currency exposure sit alongside delivery risk and are managed with financial instruments as well as project responses (PFL-AI Domains 3, 11).
- **Technology and digital.** Requirement volatility and integration risk dominate; resilience is bought through modularity, staged release and reversibility rather than buffers, so 8.4.1's levers apply in a different mix.
- **Pharmaceutical and regulated development.** Low-probability/high-consequence events and regulatory outcomes make structural treatment (8.3.1) the norm; risk files are auditable artefacts, not internal management tools.
- **Public programmes.** Optimism bias is explicitly recognised in appraisal guidance, and reference-class adjustment may be mandated; political and reputational risk carries weight that no EMV captures well.

— Case study — Domain 8: the survey Auriga did not commission (utilities)

Situation. At planning, Auriga's team considered a USD 25,000 ground survey ahead of the civil works. The register carried R2 — ground conditions worse than surveyed, p 0.50, impact USD 180,000 — and the survey was cut in a cost-reduction pass as "nice to have". The decision was not analysed; it was traded against a round-number savings target.

What happened. At week 13 the contamination was found. The reactive response cost more than the planned one would have — a second civil crew at USD 35,000, a fast-track carrying $0.20 \times 60,000 = 12,000$ of expected rework, and the exposure to a USD 45,000-per-week client penalty window that made the whole recovery urgent (Domain 6's case study; Domain 7's forecast consequence, EAC USD 4.2m against BAC 4.0m).

The analysis, done afterwards. On the decision-tree arithmetic of 8.2.3 — 0.40 probability at the time of the decision, USD 300,000 reactive versus USD 90,000 planned — the survey was worth **USD 59,000** against its USD 25,000 price. Cutting it saved 25,000 and cost the project substantially more, and the register had contained the information needed to see that.

What the domain teaches here. Two things, both uncomfortable. **A cost-reduction pass that does not consult the register is a risk decision taken blind** — savings targets and risk exposure must be traded explicitly. And **the register was not the failure; its irrelevance to the decision was** (8.3.3's test: did a decision change because of it?). The corrective adopted afterwards was procedural: any proposed saving above a threshold must state which register entries it affects and by how much.

— Case study B — Domain 8: the diversified portfolio that wasn't (technology programme)

Situation. A transformation programme's board reviewed 22 project-level risks spread across six workstreams and concluded exposure was well diversified — no single item exceeded 8 % of the aggregate EMV, and contingency was set at the aggregate mean plus a modest margin, assuming independence.

What happened. Fourteen of the 22 risks depended on the same six-person integration team. When two members left within a month, eleven risks materialised or worsened together. The aggregate outcome landed far beyond the contingency, which had been sized on an independence assumption nobody had tested — the correlated variance of 8.A.1 in practice. The programme's recovery required a re-baseline and a management-reserve release.

What was done differently. The register was restructured by **common driver** rather than by workstream, immediately revealing the concentration; the integration capability was treated as a single structural risk with a structural response (cross-training, a retained partner, staged scope) rather than as fourteen separate EMV line items; and contingency was re-derived by simulation with correlation modelled explicitly.

What the domain teaches here. Counting entries measures paperwork; counting **drivers** measures exposure. A register with many risks and few drivers is concentrated, and independence is the assumption most likely to be both convenient and wrong.

— Executive perspective — Domain 8

What a project leader cannot delegate in this domain:

- **The confidence level.** Whether contingency is set at P50 or P80 is a risk-appetite decision belonging to governance, and the leader ensures it is chosen, stated and understood — not inherited from a spreadsheet default.
- **The independence question.** Asking, of any aggregate number, which risks share a driver. It is one sentence and it is where Case study B's programme lost control.
- **The structural exposures.** Existential risks handled by structure, never by EMV — and named personally by the leader, because low probability makes them easy to leave in the register.
- **The register's relevance.** Insisting on the 8.3.3 test: which decision changed because of it this month? A register that changes nothing gets fixed or stopped.
- **The bias countermeasures.** Running the pre-mortem, commissioning independent estimates, framing continuation decisions on remaining cost and benefit — because the leader is the only person positioned to make dissent safe.

- **The AI boundary.** Detection is delegable; judgment is not, and a team that has stopped thinking because a dashboard is watching is the outcome to prevent.

— Calculation exercises — Domain 8

Exercise 8.1 Risks: A p 0.40 impact 150,000; B p 0.20 impact 500,000; C p 0.60 impact 80,000; D (opportunity) p 0.25 impact (200,000). Compute each **EMV** and the total exposure. Solution. A **60,000**; B **100,000**; C **48,000**; D **(50,000)**. Total **USD 158,000**. Common error: omitting the opportunity, giving 208,000 and overstating required contingency by 31.6 %.

Exercise 8.2 Using Exercise 8.1's four risks and assuming independence, compute σ and a P80 contingency (mean + 0.8416 σ). Solution. Variances: $0.40 \times 0.60 \times 150,000^2 = 5.40e9$; $0.20 \times 0.80 \times 500,000^2 = 40.0e9$; $0.60 \times 0.40 \times 80,000^2 = 1.536e9$; $0.25 \times 0.75 \times 200,000^2 = 7.50e9$. Total **54.436e9**; $\sigma = \text{USD } 233,315$; $P80 = 158,000 + 0.8416 \times 233,315 = \text{USD } 354,358$. Common error: taking $\sqrt{(\Sigma\sigma)}$ rather than $\sqrt{(\Sigma\text{variance})}$.

Exercise 8.3 A risk (p 0.30, impact 400,000) can be mitigated for USD 70,000, reducing probability to 0.10. Is the mitigation worthwhile? Solution. **EMV** before $0.30 \times 400,000 = 120,000$; after $0.10 \times 400,000 = 40,000$; reduction **80,000** against a cost of 70,000 $\rightarrow$ **net +USD 10,000, accept** (marginally — and the leader should note that a 400,000 impact may warrant response on survivability grounds regardless). Common error: comparing the mitigation cost against the impact rather than the **EMV** reduction.

Exercise 8.4 Two independent paths each have a 0.80 probability of meeting a milestone date and converge on it. What is the probability the milestone is met, and what is the effect called? Solution. $0.80 \times 0.80 = \mathbf{0.64}$. **Merge bias** — the merged event waits for the later path, so convergence points are systematically optimistic in deterministic CPM (8.A.2). Common error: answering 0.80, the best of the two.

— Practitioner's toolkit — Domain 8

Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.

Toolkit 8.T.1 — Risk register columns that earn their keep

ID · **cause** → **event** → **consequence** statement · affected objective · **common driver** (8.A.1) · probability · impact (cost and time) · **EMV** · qualitative band (screening only) · response family and description · **response cost and expected EMV reduction** · **secondary risks created** · owner (a person) · early-warning indicator · review date · status (open/retired/**issue**) · retirement evidence. Omit any column and the register loses a specific capability; the two most commonly missing — common driver and response cost — are the two that make it decision-useful.

Toolkit 8.T.2 — Contingency derivation sheet (one per baseline)

Register version and date · ΣEMV · variance and σ by risk · **independence assumption stated and challenged** (which risks share drivers) · method (formula approximation or simulation, with correlation treatment) · **confidence level chosen and by whom** · resulting contingency ·

management reserve separately, with its basis · draw protocol reference. A contingency figure without this sheet is a number without a defence.

Toolkit 8.T.3 — Pre-mortem session guide

Before commitment, one hour: (1) state the plan and the commitment about to be made; (2) each participant writes privately for ten minutes — "it is twelve months on and this has failed; why?"; (3) round-robin with no challenge permitted during collection; (4) cluster into causes and identify **common drivers**; (5) convert each into a three-part risk statement with an owner; (6) identify the three cheapest early-warning indicators. Rule: the most senior person speaks last, always.

— Exam preparation — Domain 8

The traps. Multiplying and summing ordinal scores as if they were money (8.2.1) · omitting opportunities from total exposure (Exercise 8.1) · setting contingency at ΣEMV and calling it funded (MCQ 8.2-D) · $\sqrt{\Sigma\sigma}$ instead of $\sqrt{\Sigma\text{variance}}$ (Exercise 8.2) · comparing response cost against impact rather than **EMV** reduction (Exercise 8.3) · managing an existential low-probability risk by its **EMV** (8.3.1) · assuming independence without testing common drivers (8.A.1) · answering the best path probability at a merge point (Exercise 8.4) · leaving occurred risks in the register as $p = 1.0$ · treating a transfer as removing exposure.

Reflection questions. 1. Take your project's contingency figure: what confidence level does it represent, who chose it, and what independence assumption sits underneath? 2. Group your register by common driver rather than by workstream. How concentrated is it actually? 3. Which decision changed last month because of your register — and if none, is the process decision support or documentation?

— Domain 8 summary

Risk management earns its place only when it changes decisions. That starts at the sentence level: cause → event → consequence, against a named objective, with opportunities identified as deliberately as threats because defensive framing otherwise buries them. Analysis then progresses from qualitative screening — useful for ordering, invalid as arithmetic — to quantification: **EMV** for each risk (Auriga's register totalling **USD 278,000**, 6.95 % of **BAC**, and USD 314,000 if the opportunity is ignored), decision trees that price choices and the **value of information** (the survey worth **USD 59,000** against its 25,000 price, and worthless if it changed no action), and aggregation to a **stated confidence** — Auriga's P80 of **USD 490,624** from a mean of 278,000 and σ of 252,642, which is defensible precisely because it says what confidence it buys, unlike a worst-case sum, an **EMV** sum or a 10 % rule of thumb. Responses are investments judged against the **EMV** reduction they buy, with secondary risks recorded and existential exposures handled structurally rather than by averaging; reserves are sized by aggregation, separated by authority, and trended against risk retirement rather than time. Beyond what can be named sits resilience — buffers, optionality, modularity, redundancy and fast detection, bought at an efficiency cost that should be chosen rather than stumbled into — together with the bias countermeasures (pre-mortem, reference-class forecasting, independent estimates) and the crisis sequence of stabilise, establish facts, decide against a clock, communicate early. Independence is the assumption most likely to be convenient and wrong: count drivers, not entries. Domain 9 turns to quality and the assurance that catches what risk analysis missed.

Index

A

AC.....	49
Accountability.....	9
Accrual.....	49
Accuracy class.....	46
Adoption.....	12
Analogous / parametric / bottom-up.....	46
Avoid / reduce / transfer / accept.....	71

B

BAC.....	46
Blended rate.....	55

C

Cause → event → consequence.....	64
Ceiling price.....	55
Commitment.....	49
Constraint.....	30
Contingency reserve.....	46
Control account.....	46
Control schedule.....	24
Convergence point.....	27
Correlation.....	66
Cost of delay.....	12
CPI SPI.....	50
Crashing / fast-tracking.....	34
Critical path.....	27
CV SV.....	50

D

Dangle.....	24
Date constraint.....	24
Daylight test.....	15
Decision tree.....	66
Displaced judgment.....	73
Draw protocol.....	71

E

EAC ETC.....	50
Early-warning indicator.....	71
EMV.....	66
ES EF LS LF.....	27
Escalation duty.....	9
Escalation of commitment.....	73

F

Feedback and delay.....	12
FFP / FPIF / CPFF / CPIF / T&M.....	55
Forward / backward pass.....	27
Free float FF.....	27
FS / SS / FF / SF.....	24

H

Hallucination.....	15
Honesty asymmetry.....	9

I

Integration milestone.....	34
Irreversibility.....	7
Issue.....	64

L

Lag / lead.....	24
Last responsible moment.....	12
Level of effort.....	50
Local optimisation.....	12

M

Management reserve.....	46
Milestone.....	30
Monte Carlo simulation.....	66

N

Named owner.....	15
------------------	----

O

Output / outcome / benefit / value.....	12
Over-trust through fluency.....	15

P

P50 / P80.....	66
Path migration.....	34
Planning package.....	30
Point of total assumption (PTA).....	55
Portfolio.....	7
Pre-mortem.....	73
Programme.....	7
Project.....	7
PV EV AC.....	50

R

Re-baselining.....	49
Recovery plan.....	34
Reference-class forecasting.....	73
Resilience	73
Resource histogram	30
Responsibility	9
Responsible-AI principle.....	7
Retention	55
Risk / opportunity.....	64
Risk appetite / threshold.....	64
Risk retirement.....	71
Rolling wave	30

S

Scenario analysis	34
Schedule hierarchy (L1-L4)	24
Secondary risk.....	71

Share ratio.....	55
Smoothing / levelling	30
Standard of care.....	9

T

TCPI.....	50
Temporary organisation.....	7
Three-point / PERT estimate	34
Total float TF	27

U

Uncertainty.....	64
------------------	----

V

VAC.....	50
Value of information.....	66
Verification proportionality	15